January/February 2014

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COVER STORY

Life-cycle and long-term migration planning

by Leif Poulsen

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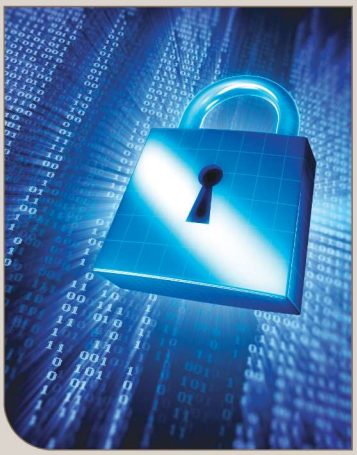


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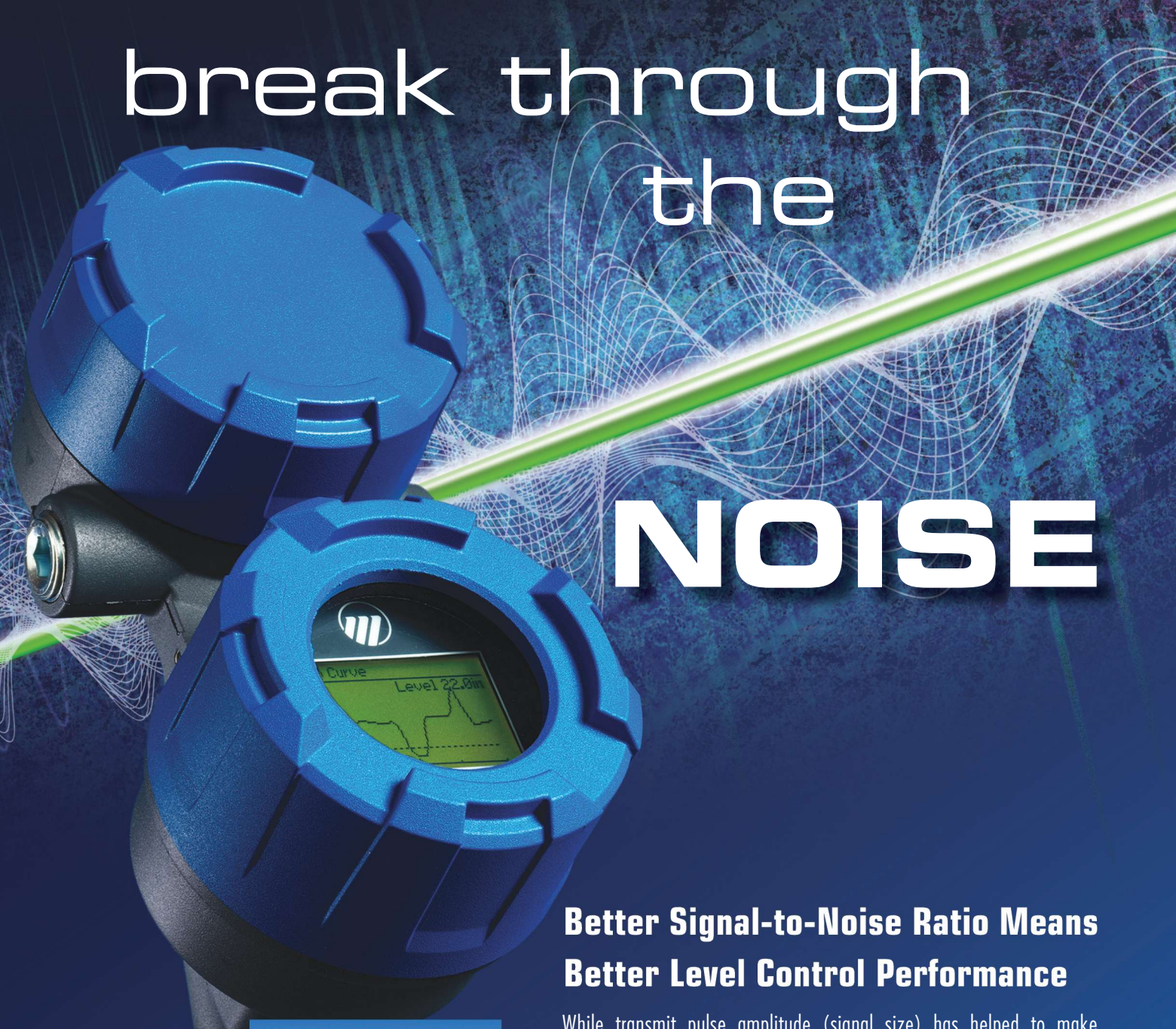
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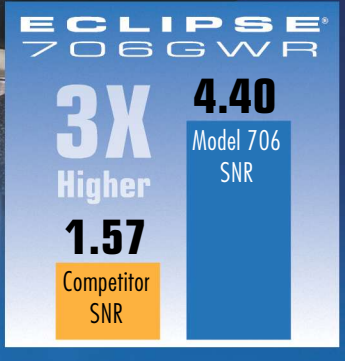
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Professionalism



By Bill Lydon, *InTech*, Chief Editor

The road to becoming an automation professional often starts as a career choice and then becomes a lifelong profession. Automation professionals are important contributors to the economy and society. They are known for their specialized knowledge. Starting with degrees and certifications as the foundation of this knowledge, they have a deep personal commitment to learn. They know that the key to quality and efficiency is professionalism, which includes skills, good judgment, and productive interaction with others.

Automation and manufacturing technology is changing at a rapid pace, and professionals are always eager to learn and improve skills. Professionals are passionate, honest, and reliable. They get the job done by finding solutions and overcoming obstacles with integrity. Professionals are the kind of people that others respect and value. In the automation profession, we are fortunate to have the International Society of Automation (ISA) as the premier professional organization since 1945. ISA sets the standards for automation and helps worldwide members and the industry solve difficult technical problems, while enhancing their leadership and personal career capabilities.

As a volunteer-driven organization, ISA depends on its members and leaders to advance the mission of the Society around the world. A great example of the high level of professionalism of ISA members was demonstrated at ISA Automation Week 2013, where people dedicated their time and talents to the automation profession. ISA members created and presented six educational tracks. Attendees benefited from the experience and know-how of leading automation and control experts, authors, innovators, and thought leaders from around the world to improve automation knowledge. Attending these sessions, the passion and commitment of participants to improve the automation industry come through loud and clear.

For decades, ISA members have been providing quality information to the automation community and creating leadership industry standards, including ISA-18, ISA-88, ISA-95, and ISA-100. Standards, such as ISA-88 and ISA-95, are now part of the fabric of industrial automation that have improved efficiency, productivity, and quality of manufacturing worldwide. This is only possible with the contributions of automation professionals participating in standards committees. In addition, ISA technical divisions provide forums for users to share ideas and best practices with activities, including symposiums, technical papers, short courses, and workshops. ISA Technical Interest Groups are professionals, aligned by common technical interests, who meet in an electronic community to share information and ideas, discuss topics of interest, share documents, and answer questions posed by other community members.

Participating in ISA, you can develop yourself as a professional by learning from worldwide subject matter experts, automation suppliers, and end users, and contribute by working together to develop and deliver the highest quality, unbiased automation information, including standards, training, publications, and certifications. ISA gives its members unparalleled access to technical information, professional development resources, and opportunities to network with other automation professionals. ISA provides many opportunities to network with peers in your industry and offers forum discussions on industry-wide challenges where subject matter experts share ideas and solutions. ISA members are advancing the state of the art and developing the automation profession.

In your journey to become an automation professional, consider joining ISA and contributing your time and talents to advance the automation industry. You will find a community of people with common interests, challenges, and ideals.

ISA *INTECH* STAFF

CHIEF EDITOR
Bill Lydon
blydon@isa.org

PUBLISHER
Susan Colwell
scolwell@isa.org

PRODUCTION EDITOR
Lynne Franke
lfranke@isa.org

ART DIRECTOR
Colleen Casper
ccasper@isa.org

SENIOR GRAPHIC DESIGNER
Pam King
pking@isa.org

GRAPHIC DESIGNER
Lisa Starck
lstarck@isa.org

ISA PRESIDENT
Peggie W. Koon, Ph.D.

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Modular construction

I agree fieldbus is good for modular construction, which is not just for floating production storage and offloading (FPSO) [*InTech* July/August 2013 Web Exclusive]. Projects in the process industries are increasingly turning to modularized construction to reduce escalating construction costs and to use construction resources where they are available instead of where they are scarce. Fieldbus and modularization are industry trends seen for ships such as FPSO and floating liquefied natural gas floaters, but also for onshore plants everywhere from oil and gas to metals and mining—on the even larger scale of mammoth modules.

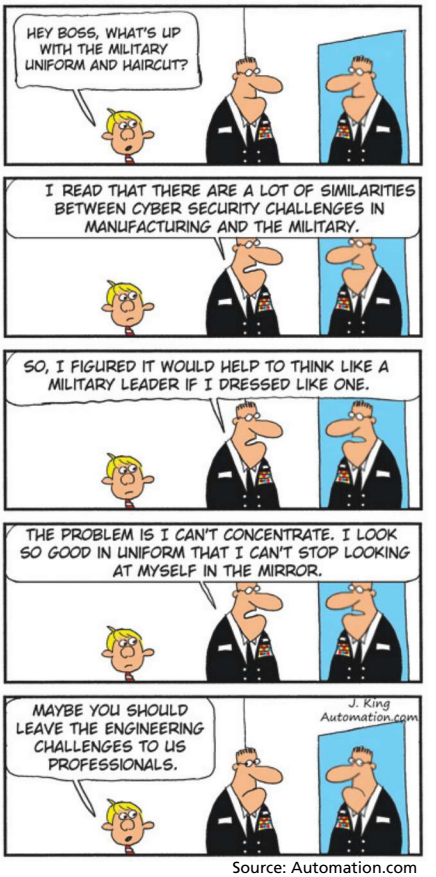
Fieldbus fits well with modularization, because the instrumentation can be tested on the module in the yard and when the module arrives at the site. In the yard, all instrumentation is installed and hooked up on the module, fully commissioned and tested. Thanks to the digital bus architecture, fieldbus reduces the amount of tray to install, cable to lay, and wires to cut, strip, label, ferrule, and connect on the modules. Device commissioning is faster. The traditional five-point loop check associated with 4–20 mA is not required. Once a module arrives on site, minimal wiring has to be connected. System integration is simply a matter of plugging the module into the established power and fieldbus leading back to the system in the equipment room. Fieldbus makes the instrumentation and control (I&C) part of the modules easier to assemble, reducing the I&C connections to a minimum. Because the "topsides" consists of process modules fabricated in different yards, fieldbus enables these modules to be connected with the rest of the ship using only a few cables; this is a "plug-and-play" concept ideal for fast-track projects.

Using fieldbus has many secondary benefits, one of which is the ability to automatically monitor device diagnostics from a central location. Device diagnostics from the intelligent device management software part of the asset management solution mean that physical inspection of the device is rarely required. Subsequently, devices do not need to be installed in conveniently accessible locations, but can instead be directly mounted, on the pipe or equipment, without impulse lines. This simplifies installation on compact modules and also eliminates the risk of plugged impulse lines.

Another benefit of fieldbus is the ability to place analyzers on the module in the field or on deck, connected directly to the process sampling point, without having to run long sampling lines with heat tracking off the module to an analyzer shelter. This may include gas chromatographs and other analyzers. This greatly facilitates the completion and integration of modules at the site. The analyzer shelter may not be required, reducing weight—important on offshore vessels and installations. Note that these are field-mount analyzers in their own suitable enclosure. The analyzers connect to a fieldbus running back to the distributed control system and AMADAS.

Fieldbus eliminates the problems associated with hardwired I/O and is an ideal fit for modular construction. The reduced footprint and weight is important on a crammed FPSO. As devices these days are no longer integrated with just one signal each, the system I/O counts are higher than in years gone by. But a fieldbus system is designed based on device count instead of I/O count, so 6,000 I/O instead becomes 2,000 devices.

Jonas Berge



Source: Automation.com

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ISA introduces cybersecurity certificate program

ISA's new certificate program, the ISA99/IEC 62443 Cybersecurity Fundamentals Specialist Certificate, is designed to help professionals involved in information technology and control systems security improve their understanding of ISA99/IEC 62443 principles and acquire a command of industrial cybersecurity terminology.

A cross section of international cybersecurity subject-matter experts from industry, government, and academia developed the series of ISA99/IEC 62443 standards. The standards apply to all key industry sectors and critical infrastructure, with the flexibility to address and mitigate current and future vulnerabilities in industrial automation and control systems. The certificate will be awarded to those who successfully complete a two-day ISA classroom training course, *Using the ANSI/ISA99 (IEC 62443) Standards to Secure Your Industrial Control System (IC32)*, and pass a 75-question, multiple-choice exam.

Dick Morley joins Memex Automation board

Memex Automation named inventor, entrepreneur, and angel investor Richard E. ("Dick") Morley to the company's board of directors. He won the Business Week Entrepreneur of the Year Award in 1990 and is the namesake of the Richard E. Morley Society of Manufacturing Engineers Outstanding Young Manufacturing Engineer Award. When asked why he joined the company as a director, Morley said, "It is about time that CNC [computer numerical control] machines join the concept of supervisory control and data acquisition in an efficient manner. This team brings the tools, technology, and understanding to connect every machine to management in real-time, showing them how to make more money."

Siemens introduces Managed Security Service

Siemens Industry announced its Managed Security Service aimed at providing continuous protection to production environments. The service assesses security posture, implements recommended security measures, and transitions into ongoing defense against cybersecurity threats in industrial control system environments.

The Industrial Security Services group expands the existing Siemens security portfolio by providing holistic protection to manufacturing sites. Siemens' approach is to partner with customers to help them build sustainable industrial security programs and move away from point solutions for security to a comprehensive security program delivered through a managed security service.

Fieldbus Foundation updates specifications

The Fieldbus Foundation announced updates to its open, nonproprietary FOUNDATION fieldbus technical specifications. The new features will enhance the usability of FOUNDATION technology, with the ultimate goal of making the digital fieldbus automation experience easier to use than conventional analog control systems.

The updated version of the FOUNDATION specification includes additional advanced support for fieldbus device replacement and backward compatibility, device description (DD) templates, field diagnostics, and alarm/alert integration. The ability to use these enhanced features is included in the foundation's new Host Profile C, which will be available for registration in early 2014. The latest features enable host and device suppliers to offer backward compatibility to their users to further simplify device replacement. The incorporation of DD templates is an interoperable way for instrumentation companies to offer a variety of application settings for devices. Suppliers can embed the templates in their DDs from the factory floor. This will make it quicker and easier for end users to access a correct offline configuration.

ASM publishes second edition of HMI Design Guidelines

The Abnormal Situation Management (ASM) Consortium released a second edition of its display design guidelines under the new name *Effective Console Operator HMI Design Practices*. This best practices guidelines book is based on the ASM Consortium's many years of research in preventing, mitigating, and managing abnormal situations. The book is written for individuals who establish company human-machine interface (HMI) standards, style guides, and information displays accessed by console operators through their control system workstations. This audience includes individuals from both manufacturing and vendor companies who deliver solutions for console operator workstations.



The new edition has consolidated the number of guidelines to 64, compared to 81 guidelines in the previous edition, and has revised more than 50 percent of the guidelines. In addition, the new edition contains a new section on HMI design philosophy and a new appendix summarizing results from an HMI design case study.

Endress+Hauser appoints Matthias Altendorf CEO

After 19 years at the top of Endress+Hauser, Klaus Endress moves to the group's supervisory board as of 1 January 2014. He replaces President Klaus Riemenschneider, who is retiring after 43 years with Endress+Hauser. The new CEO of the group is Matthias Altendorf, former managing director of the Center of Competence for level and pressure measurement engineering in Maulburg, Germany. He is only the third CEO in the company's history, which started in 1953, and the first one not coming from the Endress shareholder family.

Automation by the Numbers

3 million

The Chinese market for industrial Ethernet and fieldbus technologies grew by 18 million nodes in 2012. More than **3 million** nodes used Ethernet, and the remainder used fieldbus technology. Although fieldbus has a large base of new connected nodes in China, it is not as commonly used as in developed countries such as Germany or the U.S. This is mainly because Chinese users are encountering networking technology much later than those in developing countries.

However, the growing speed of Ethernet is considerable in China; there is a great opportunity for Chinese companies to upgrade their automation systems under current market conditions. Companies can jump from old fieldbus technologies directly to Ethernet, as many are now doing. The Chinese market is currently engaged in extensive upgrading and new infrastructure construction, and that will require many Ethernet applications.

In China, international brands are influential. This is also true for industrial networking protocols, because most of them have supporting companies. For example, the most popular fieldbus protocols in China are PROFIBUS and CC-Link, which are developed and promoted by Siemens and Mitsubishi separately. Some open protocols also have a large number of nodes connected; the most representative ones are CANOpen, Modbus, and HART. However, these three protocols do not deliver strong functionality and are more likely to be used in low-end applications for easy connections.



With the upgrading and construction in China, companies, including industrial automation vendors, are also being compelled to upgrade their systems using Ethernet. Most protocols have Ethernet variants. Because of this, many fieldbus users will turn to the Ethernet of the application, for example, PROFIBUS to PROFINET, CC-Link to CC-Link IE. The new automation products will also support those new Ethernet connections.

760,000



According to Berg Insight, the shipments of cellular M2M devices in industrial automation reached **760,000** worldwide in 2013. With a compound annual growth rate of 22.5 percent, shipments are expected to reach 2.1 million in 2018. The market is served by players with varying backgrounds. Eaton, Phoenix Contact, Advantech, and Kontron are major providers of industrial automation equipment and are also important vendors of products and solutions featuring embedded cellular connectivity. Industrial network equipment specialists such as Moxa, Westermo, and B&B Electronics are also major vendors of cellular solutions. Other significant vendors include M2M specialists such as Digi International, Calamp, Maestro Wireless, and Viola Systems. Netmodule and eWon are examples of companies with highly specialized offerings targeting the industrial automation industry.

Backbone network communication and remote monitoring are the two largest applications for cellular M2M connectivity within industrial automation. Remote service maintenance and diagnostics of machinery and industrial robots is a major application within factory automation, and real-time monitoring of remote facilities and equipment is one of the most common applications within process automation.

$7.2 billion

Koch Industries acquired Molex for **$7.2 billion.** The acquisition was finalized through the merger of Koch Industries' wholly owned subsidiary, Koch Connectors, Inc., with Molex. As a result of the merger, Molex is now an indirect wholly owned subsidiary of Koch Industries, Inc., retaining its name and headquarters in Lisle, Ill. The current management team will continue to operate the company. Based in Wichita, Kan., Koch Industries, Inc., is one of the largest private companies in the U.S. with annual revenues of about $115 billion. It owns a diverse group of companies. Molex is a 75-year-old global manufacturer of electronic, electrical, and fiber-optic interconnection systems.

$908.7 million

The growing demand for customized systems for critical oil and gas applications is brightening the prospects of the European power conversion market in the oil and gas industry. Escalating energy costs and exploration of new oil and gas fields also fuel the uptake of power conversion solutions in the region.

Frost & Sullivan found that the market earned revenues of **$908.7 million** in 2012 and estimates this to reach $1,151.9 million in 2017. The research covers electric drives and electric motors, with the latter cornering 78.9 percent of the market share. However, electric drives have the higher growth rates due to their high-tech, energy-saving functions and ability to decrease downtime costs through control and efficient rotating assets. They also can reduce the costs of upstream, midstream, and downstream activities.



15 MW

ABB commissioned four low-speed dual-pinion drive systems at the Detour Lake gold mine in Ontario, Canada. These systems include the largest such mill drives in the world. The low-speed drives operate without a gearbox, with motors driving the pinions directly, increasing overall system efficiency.

ABB's scope of supply included four systems for two semiautogenous and two ball mills, each consisting of two synchronous motors, converter transformers, and ACS6000 frequency converters. This includes active rectifier units that allow the mill drives to achieve a power factor of 0.90 leading. ABB also delivered the programmable logic controller to control the drives and mill auxiliaries. Each mill drive has a rated power of **15 MW.** ABB's drive systems have variable-speed operation, real-time frozen charge protection, and the frozen charge remover function.



Life-cycle and long-term migration planning

Successfully upgrading and replacing systems in a running production environment

by Leif Poulsen

Automation and manufacturing information technology (IT) systems are characterized by a relatively short life. Often the systems have to be upgraded or replaced long before the process equipment has reached the end of its life cycle. For most companies, it is a challenge to manage upgrading or replacing systems in a running production environment, and often the need for upgrade or replacement is ignored until it appears as an unpleasant surprise. This article shows how it can be done successfully with careful planning and organization.

Two main factors drive the need to upgrade or replace automation and manufacturing IT systems: the technical deterioration of the automation and manufacturing IT systems and changes to the requirements of the business processes such systems support.

The reliability of technical systems will decrease over time if companies ignore migration activities such as upgrading operating systems, database systems, and application software. The operational risk of failure increases accordingly. With careful planning, the operational risk can be kept at an acceptable level, while protecting existing investments and minimizing life-cycle cost. For a typical automation/IT system, only 20 to 40 percent of the investment is actually spent on purchasing the system; the other 60 to 80 percent goes toward maintaining high availability and adjusting the system to changing needs during its life span.

Along with assessing the necessary migration activities to cope with technical deterioration, it is also important to assess new challenges and new opportunities from a business point of view. The business environment is constantly changing, and opportunities for improving existing or exploiting new technologies must be considered. Typical business objectives, which may be important business drivers for migration planning, include speed to market, competitiveness, growth, quality, and compliance.

Long-term migration plan

Creating a long-term migration plan helps companies to keep system operational risk at acceptable levels while meeting changing business requirements. A migration plan addresses risk mitigation and timely support of business goals. It takes into account important constraints such as current Good Manufacturing Practice (cGMP) compliance, technology functionality and performance, system support, and plant downtime.

Drawing the future system landscape by making

all these goals and constraints add up with minimum investment is the key to a good migration plan. Together with a phased and robust implementation plan, this ensures a painless transition.

The overall approach to long-term migration planning is illustrated in figure 1. The migration plan takes an organization to where it wants to be in five years by steps that match the needed changes with the resources available to implement the changes. This approach is based on architectural design principles as defined in The Open Group Architecture Framework (TOGAF) standard, which is widely accepted for developing enterprise architectures.

We distinguish between the current architecture and the target architecture, which correspond to the description of where the company is now and where it wants to be, respectively. The migration plan goes from the current to the target architecture, maybe via some temporary transition architectures.

Each architecture must be described at a number of layers with appropriate mapping between business and technology, as shown in figure 1. We operate with the following layers:

- Business objectives are part of the overall strategy work. This is valuable for setting the proper direction of the planning process.
- The business model provides context to understand manufacturing and business processes. It normally includes a high-level description of the material/process flow.
- Describing manufacturing and business processes is essential to successfully applying technology and accurately estimating business value.
- Information, data, and documents are essential to linking processes and applications. The main concern is transactions and orchestrating information flows between applications.
- Application descriptions form high-level requirements and define interfaces.
- Define infrastructure, computers, and networks requirements (hardware, availability, and performance).
- Enabling services define how to secure efficient and successful operational control and support of the solutions.

Developing the migration plan

The development of the migration plan for a complete organization, site, or single facility may be a complex task that involves a lot of people. It is recommended that groups organize the development task in the five steps briefly described below.

FAST FORWARD

- Automation and manufacturing IT systems often have to be upgraded or replaced long before the process equipment has reached the end of its life cycle.
- Technical deterioration and new business process requirements drive the upgrade or replacement of automation and manufacturing IT systems.
- Concepts described in this article explain how to successfully upgrade or replace automation and manufacturing IT systems.

Step 1: Mobilize

The purpose of the mobilize step is:

- to get a common understanding of objectives and goals
- to mobilize the project organization
- to detail the consulting plan, milestones, and deliverables
- to gather available inputs
- to ensure a proper understanding of concepts, practices, and theory

During this step, the following activities are conducted:

- planning meetings
- kick-off workshop

The output is:

- detailed consulting plan
- overall goals
- process overview

Step 2: Analyze

The purpose of the analyze step is:

- to analyze the business and manufacturing processes in order to
  - assess readiness for automation and manufacturing IT support
  - clarify data and functionality needs for to-be architecture
  - identify key benefits for setting goals and business justification
- to establish an as-is architecture
  - existing manufacturing processes with possible relation to automation, data collection,

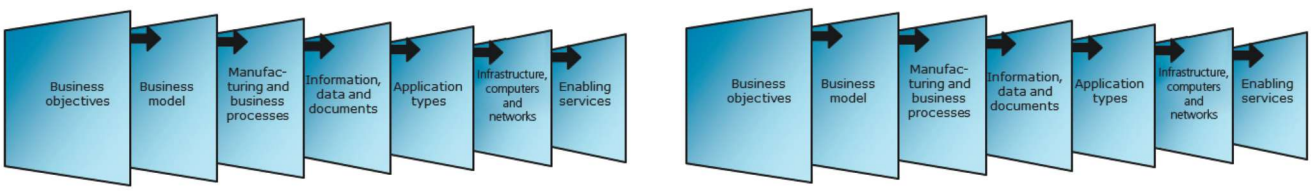


Figure 1. Overall approach to development of long-term migration plan (based on TOGAF)

and manufacturing execution systems
  - existing business processes with possible relation to manufacturing systems
  - existing data
  - existing applications and interfaces
  - existing logical and physical infrastructure
  - existing support services

During the analyze step, the following activities are conducted:
- process interviews and workshops
- plant tour to get information in context
- as-is systems and infrastructure architecture interviews and workshops
- scoring enabling services compliance and maturity

The output of this step is:
- as-is architecture
- analysis documentation
- ideas list of challenges and opportunities

Step 3: Target

The purpose of the target step is to identify the needs determined in the analyze step and describe the targets. The solution or target architecture that is the goal will describe:
- to-be business processes and functionalities
- target application types with supported overall functionality, users, information, and interfaces
- infrastructure needs and the revised supporting services

During the target step, the following activities are conducted:
- process improvement interviews and workshops
- architecture improvement interviews and workshops

The output of this step is:
- to-be architecture (presentation)
- short-description of application types

Step 4: Justify

The purpose of the justify step is to make preliminary business justifications based on rough cost and benefit estimation.

The gap between the as-is and the to-be leads into a number of project ideas. A justification of the collected ideas shall be elaborated to separate need-to-have from nice-to-have and to present and review project ideas with management.

During the justify step, the following activities are conducted:
- rough cost and benefit evaluation
- draft to-be presentation

The output of this step is:
- overall goals summary
- business case/ideas prioritization
- resource need

Step 5: Plan

The purpose of this step is to plan execution based on priority, resources, and dependencies:
- to plan the sequence in execution of the consolidated project portfolio
- to ensure resources and competencies for the next steps
- to initiate governance activities
- to complete the consultancy and hand over deliverables

During this step, the following activities are conducted:
- implementation planning
- investment planning
- risk assessment

The output of the plan step is:
- implementation plan
- employee load profile adhering to the plan
- project risk assessments
- rough cut investment plan
- final presentation

Practical example

The following example shows how to apply the described approach in real life. For confidentiality, all data in the case has been made anonymous. However, it is about a fairly large site producing active ingredients for a number of pharmaceutical products. The site was established more than 20 years ago, and although some equipment and system upgrades have been made since then, a number of outdated systems still need to be replaced. The building management systems and the distributed control systems (DCS) are especially based on old technologies, which are now hard to support. Now the site also has to adapt to new business requirements, including retiring some products and launching new products. So there is an overall need to look at a migration plan that covers both technical and business requirements.

First, it is important to create an overview of what is currently installed in the different facilities. This information is often hidden in a huge number of documents (and maybe also in the minds of people) and has to be extracted and visualized to be the basis for the migration planning. To do this, we typically develop a Process Module Diagram, which shows the main equipment and the material flow in each facility. As separate layers on top of this drawing, we illustrate which systems are supporting which equipment. An example is shown in figure 2. The data about the installed systems is also recorded in a system repository (or can simply be saved in an Excel spreadsheet), which can be used for further analysis and planning.

Before discussing the migration planning, it is important to identify the main business drivers for the changes on the site. In this case, site management stated the main business drivers as follows:
1. right first time, compliance
2. time to market, flexibility
3. sustain success, competitiveness,

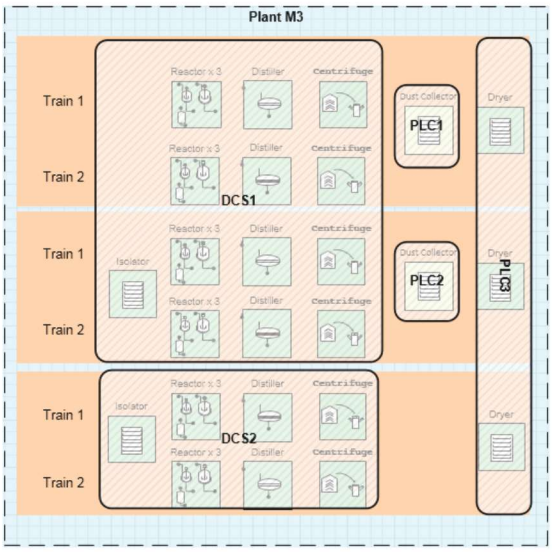


Figure 2. Process module diagram with automation layer to show current system installations

operational excellence
4. right first time, quality
5. volume growth

Such business drivers have to be turned into more specific business objectives that can be measured. We need this to prioritize potential change projects.

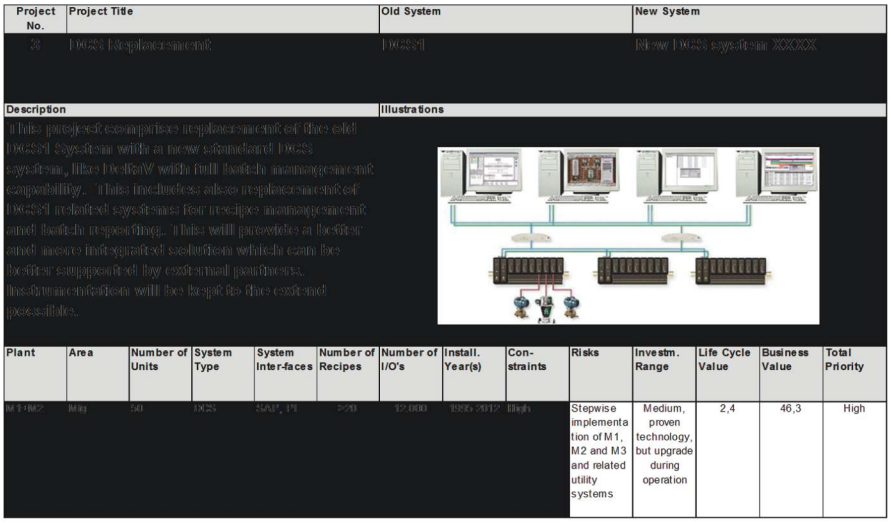
Next, we need to know more about how well the existing systems support existing and future business processes. We use a standardized reference model (based on the ANSI/ISA-95 series of standards) to explore this. It comprises 19 high-level business processes that are broken down to the relevant level of details to understand weaknesses and the need for changes from a business point of view.

In addition, we also have to assess the technical capabilities of the existing systems to support the business processes in the future. This is done systematically with the information described above in the system repository. For each system in the repository (in this case about 70 systems), the following aspects must be assessed:

- hardware condition (history of failure or mean time between failure, age of equipment, availability of spare parts)
- software condition (support from vendors, availability of documentation, availability of competences)
- system restore capability (redundancy, mean time to repair)
- business impact assessment (disclosure of information, data errors, non-availability)
- indicative scoring (system reliability, system criticality, and life-cycle management score)

The technical assessment reveals a need to upgrade and replace a number of systems:

- The process control systems are based on one common outdated DCS platform and many different programmable logic controllers, of which some are also ready for replacement.
- The building management system is based on a newer platform but needs an upgrade to fit new requirements.
- A number of the support systems also need some upgrades, and some need to be replaced.
- The infrastructure that is common for all systems needs to be better segmented and protected to meet current security requirements.



Project No.	Project Title	Old System	New System
3	DCS Replacement	DCS1	New DCS system XXXX

Description	Illustrations
This project comprises replacement of the old DCS1 System with a new standard DCS system, like DeltaV with full Batch management capability. This includes also replacement of DCS1 related systems for recipe management and batch reporting. This will provide a better and more integrated solution which can be better supported by external partners. Instrumentation will be kept to the extend possible.	

Plant	Area	Number of Units	System Type	System Inter-faces	Number of Recipes	Number of I/O's	Install. Year(s)	Con-straints	Risks	Investm. Range	Life Cycle Value	Business Value	Total Priority
M1+M2	Mix	50	DCS	SAP, PI	>20	12.000	1995-2012	High	Stepwise implementation of M1, M2 and M3 and related utility systems	Medium, proven technology, but upgrade during operation	2,4	46,3	High

Figure 3. One-page description of potential migration project

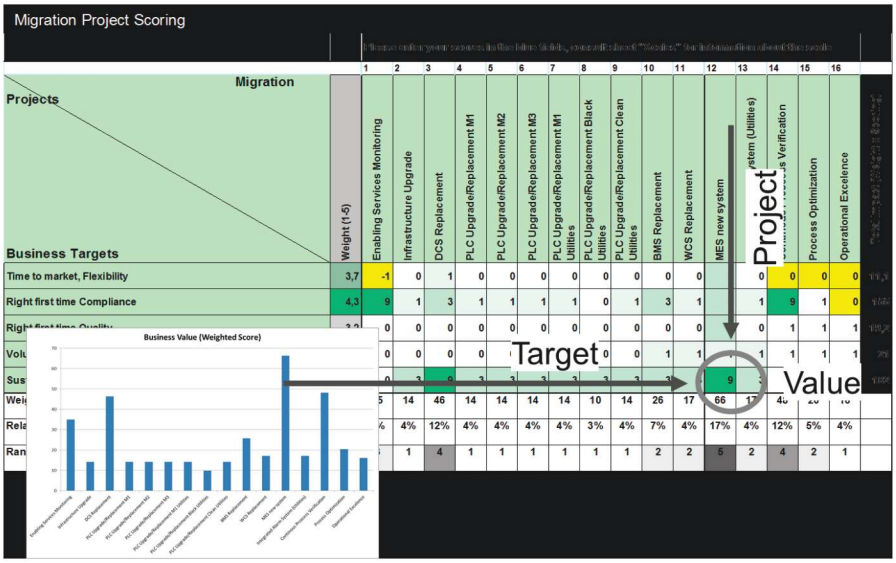


Figure 4. High-level business impact assessment of migration projects

By looking at the business objectives for the future, it is evident that none of the existing systems can fully cope with the future requirements. This knowledge led to several ideas for introducing new technology, including a new manufacturing execution system. The analysis ended up proposing 16 different projects, which can be implemented in steps to meet both technical and commercial requirements.

The technical scope and the cost of each project are estimated, and a one pager for management discussion is prepared for each project proposal (see example in figure 3).

To prioritize among the proposed projects, an impact assessment is made for each project. The impact assessment includes both an assessment of the impact on the business objectives and an assessment of the impact on the reliability of the systems.

Figure 4 shows how the various projects affect the main business objectives, and figure 5 shows how the projects affect the reliability of the systems.

The next step is to refine the budgets for the selected projects (figure 6) and to schedule the sequence for execution according to the priorities and

technical constraints.

Normally you have to simulate a number of implementation scenarios to assess the overall need for resources and capital for each project per period in the long-term migration plan (figure 7). One of the major constraints to consider is opening windows in the production schedule where you can modify or replace the systems. Often these windows are limited to holiday shutdowns, and this may be a bottleneck for execution.

Because the time for the change and cut over to new systems normally is very limited, you have to prepare thoroughly. Everything has to be planned at the necessary level of detail. An important aspect of the planning is the validation of the upgraded systems; good advice about this is provided in reference 3. In this case, the execution of the long-term migration plan has been organized in six separate implementation programs, as shown in figure 8.

Part of the preparation is also careful assessment and mitigation of project risk. Figure 9 shows typical risks associated with migration projects.

Business-driven process

The approach to life-cycle management and long-term migration planning described in this article is business driven. It includes an assessment of current and future business objectives and a careful analysis of how the technical system must be maintained, extended, or replaced to best support these objectives. Further, the approach is based on system architectural design principles (TOGAF), which allows for stepwise implementation according to the availability of investment budgets and qualified resources. It includes a description and assessment of the current and future system architectures as key elements in identifying relevant migration projects. Finally, the approach is based on organizational change management principles, which ensure timely involvement of relevant stakeholders to make the implementation of the migration projects successful. The feasibility of this approach has been demonstrated through many practical examples, as described in this article.

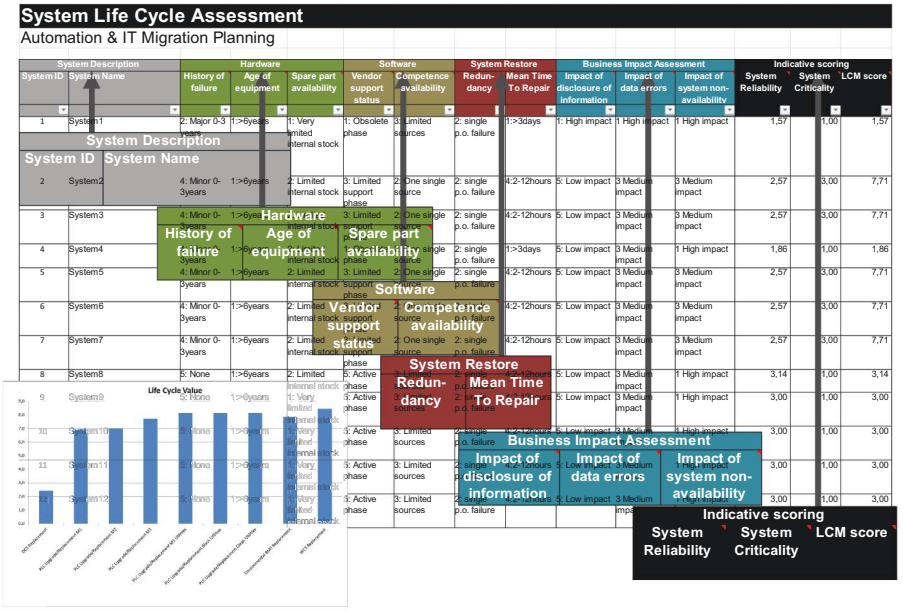


Figure 5. High-level reliability impact assessment of migration projects

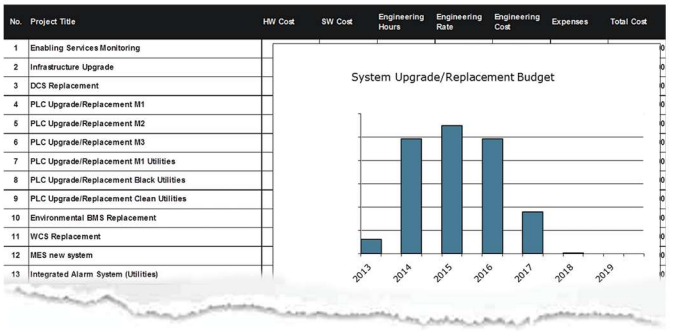


Figure 6. Consolidated overview of migration budget (actual values not shown for confidentiality)

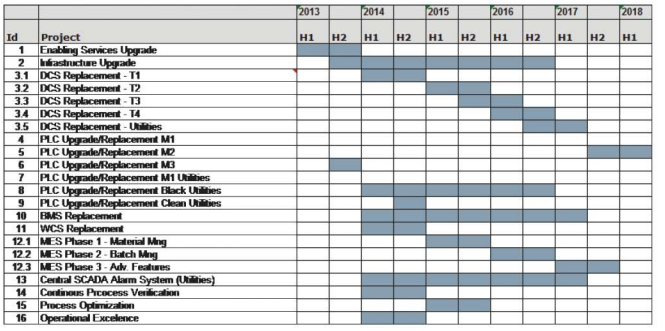


Figure 7. Consolidated overview of migration schedule

ABOUT THE AUTHOR

Leif Poulsen (lpou@nnepharmaplan.com), senior specialist, automation and IT, at NNE Pharmaplan, holds a master's and a Ph.D. in process engineering and is certified as a professional enterprise architect according to the TOGAF 9 standards. At NNE Pharmaplan, Poulsen is responsible for the development of technology, methods, and competencies within automation and IT and works as a senior business consultant for customers worldwide. He is an expert on business analysis and conceptual design of automation and IT solutions, including how to deploy such solutions effectively in a GxP regulated organization. He is an active member of the ISA88 and ISA95 standards committees.

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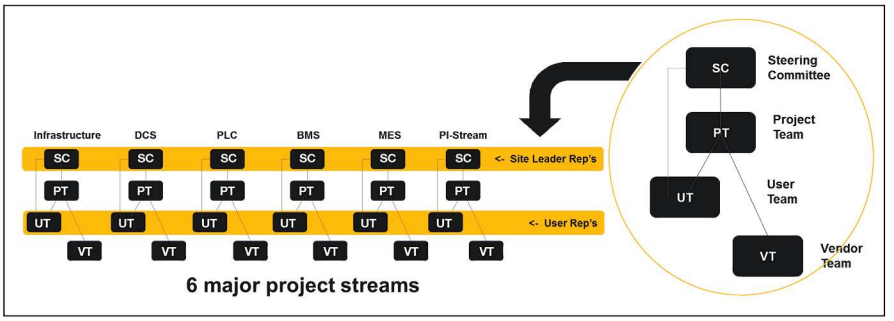


Figure 8. Organization of migration projects in six streams

1. Availability/bottleneck of skilled resources (development, operation, support)
2. Complexity higher than expected (unclear requirements)
3. Scope creep – impact on budget and schedule
4. Quality of documentation of existing systems
5. Interfaces to and impact on other processes and systems
6. Alignment with other corporate and site projects and standards (governance)
7. Management prioritization and support

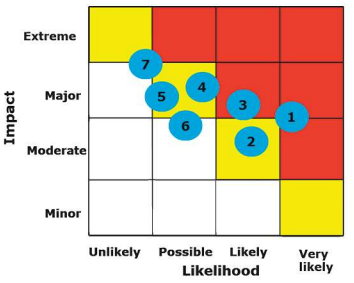


Figure 9. Assessment of typical risks by migration projects

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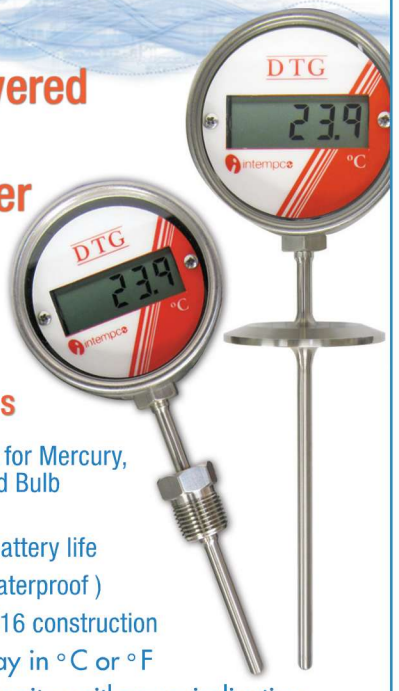
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Operator training simulators in the modern plant

Operator training simulators raise operator competency, improving plant performance and reliability

By Peter Richmond

With as much as 40 percent of the current workforce in some industries retiring in the next five years and increasing difficulty in attracting new talent, innovative training and certification solutions are essential. Keeping plants operating safely, with optimal performance and reliability, requires companies to find new ways to improve the competency of their operators. This article describes some of the promising technologies that will shape operator training in the coming years.

Operations staff directly affects plant safety, availability, and reliability. Rising fines and costs associated with incidents such as unplanned outages, accidents, and spills are among the additional factors driving the creation of new programs for safety and operational excellence. Operator training simulators (OTS) are a key component of such initiatives. Process manufacturers buy simulator systems to train new employees, to update the skills of existing engineers, and to institutionalize and retain the knowledge of experienced operators. Although dynamic simulation and operator training simulators have been available for a long time, technology and applications continue to evolve around customers' growing needs to improve training. Now companies are investing in three areas: integrating OTS systems into conventional corporate training programs, running OTS in virtual environments to avoid project schedule conflicts, and adding three-dimensional (3-D) virtual reality capability to reduce training costs and improve safety.

Integrating OTS

Course development, training materials, and instruction are the fundamentals of any good OTS program. However, a modern environment also requires companies to deliver traditional programs in different ways and to integrate the information with existing learning management systems (LMS). Over the past three or four years, there has been a growing interest in the concept of corporate operator training simulators (C-OTS) that address this requirement, including increased adoption of the sharable content object reference model standard, which facilitates sharing among disparate e-learning systems.

FAST FORWARD

- Corporate operator training simulators enable companies to share training materials globally.
- By developing OTS systems on virtual servers, organizations can start training well before system commissioning.
- Three-dimensional virtual reality simulation is an engaging, high-fidelity, safe, and cost-efficient route to hands-on experience.

Conceptually, a C-OTS encapsulates a remotely hosted operator training simulator. The solution involves a client-server architecture in which the server runs the simulation model remotely, and the customer accesses the simulator through the client.

Maximizing the benefit of such simulators requires an organized, comprehensive training program that optimizes the use of the simulator and operator job performance. Doing so effectively requires upfront attention to details and answers to the following questions:

- How will I conduct training on the OTS once it is delivered?
- How will the OTS be incorporated into the programs we already have in place?
- How can I leverage the plant data and training materials we already have in producing my simulator training materials?
- How do we make sure the investment made in an OTS has long-term value?

Addressing these questions early in the process can add significant value to the OTS investment and, if a comprehensive plan is built around such questions, can safeguard that value for the life of the plant (while also capturing the knowledge and experience associated with the simulator construction and validation). The five-phase, performance-based training approach shown in the diagram is an effective guide to reaching this goal.

Following this procedure will help companies develop a comprehensive, translatable, performance-based program that meets the operational goals of the facility. Trainees would learn to respond to upsets in proper, predictable ways, and their growth as operators would become real, visible, and documented.

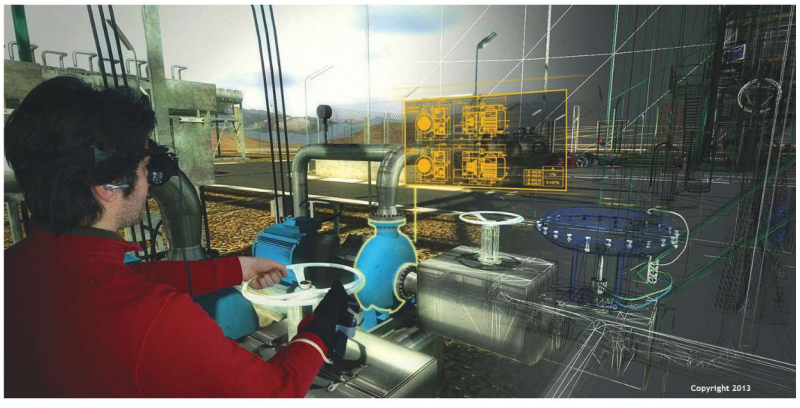
Control and safety testing

Concurrent development of control and safety systems can sometimes require training to begin before the system is commissioned. In an ideal world, there would be plenty of time first to develop all the controls, then test the controls thoroughly, integrate them with the process model, test the entire system as a whole, and finally turn the simulator over to the client for operator training—all well before the commissioning of the plant, preferably three-to-six months in advance. In reality, however, many OTS projects struggle to meet this schedule, often because of late changes to plant and control designs from the engineering procurement and construction firm.

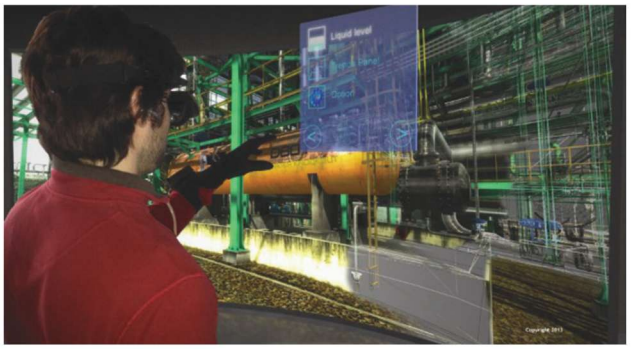
Once all parties understand such conflicts, they can address them through informed, coordinated project management and smart engineering productivity tools. Separating the control algorithms from the real hardware allows the algorithms to run fast, slow, or in a single-step mode to facilitate validation and training. A modern simulation environment discussed earlier, for example, includes a wide range of sophisticated tools to allow concurrent development of the operator training simulator and the development, testing, and retesting of the control and safety system.

Hardware virtualization is another area in which technology can support the delivery of complex integrated projects and the development of the OTS system itself. High-fidelity operator training simulators often include relatively complex hardware architectures with multiple workstations required for the various simulation, controls emulation, and operator workstation components. Implementing several OTS systems for numerous key plant units multiplies this hardware real estate challenge. Developing the systems on virtual servers allows the hardware and operating system to be decoupled from the software applications needed to train personnel or simulate a plant control system.

Deploying virtual images on machines capable of serving multiple applications to multiple users can have many advantages. Not only might this reduce the hardware footprint of the installed system, it also gives more flexibility



Improving safety-critical skills by enabling operators to perform tasks in a simulated environment, allowing them to react quickly and correctly, facilitating reactions in high-stress conditions, and instilling confidence and standards for teamwork and communications.



C-OTS system using Invensys DYNSIM dynamic simulation software is accessed through a locally installed or cloud-based portal fully integrated with the simulator.

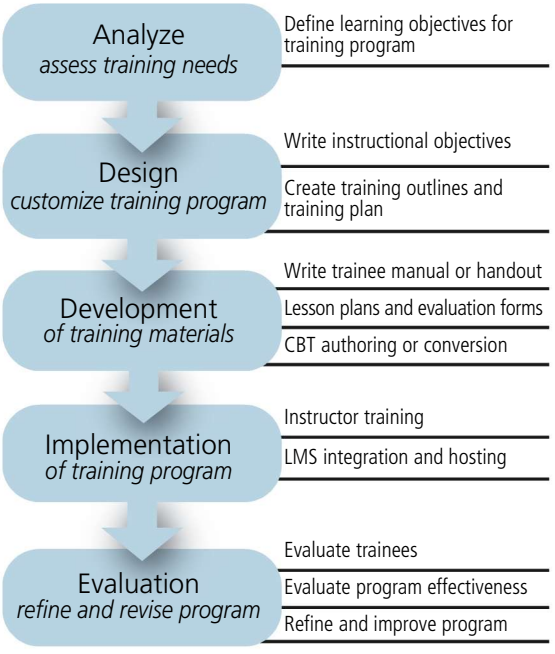
in how the OTS is engineered and utilized during the control testing phase of a project, as well as how it is maintained and supported throughout the life of the plant.

3-D virtual reality training

Once reserved for cutting-edge engineering and creative industries, 3-D visualization is being used in new and innovative ways across a number of industrial sectors, helping to safely and effectively train plant operators and staff.

The emergence of 3-D visualization as a method of training has grown out of the need of many industrial companies and organizations to instruct their employees in a safe and secure environment. In addition, the nature of continuous process plants necessitates minimal downtime, and there are often few opportunities for initial training of new staff and for ongoing training for experienced staff, particularly in scenarios that may only occur rarely.

Plant operators can benefit from incorporating 3-D visualization into their training systems in many ways. Chief among these is the ability to have high-fidelity operations, maintenance, and safety training in a cost-effective, low-risk setting. Putting people in the field in dangerous and often remote locations, such as offshore energy platforms, strictly for training purposes, is not only costly, but also risky to platform operators, their co-workers, the facility itself, and the environment. Because of advances in simulation, visualization, and interactive gaming technology, it is now possible for offshore operators to learn much of their craft in a safe, realistic training environment.



Five-phase, performance-based training approach

Virtual reality simulation is particularly well-suited for industrial training, where remote, unsafe, and pressure-filled sites are increasingly common. This type of technology enables platform operators to receive a large portion of their training in a virtual environment, reducing cost and risk. For example, risk of injury can be eliminated because operators are not immediately placed in an unfamiliar environment. After going through such training programs, operators are less likely to make mistakes such as spills or shutdowns, which could have serious consequences. They are also less likely to encounter emergencies they have never before experienced. In situations where units are shut down only once every year or less often, virtual reality training is an invaluable practice tool for staff at all levels.

Many believe this approach fits especially well with the new generation of engineers and plant operators who are already familiar with this technology and who are used to an entirely different learning environment than previous generations experienced. In addition to providing a more realistic training environment, 3-D virtualization training ensures a more interactive and hands-on experience.

It is a very exciting time for many industrial companies and organizations as they help drive virtual reality solutions and create innovative and practical applications directly relevant to their staff's needs. As the conditions and demands on the industry evolve, plants of all kinds are increasingly using virtual environments to help plant operators and staff rapidly adapt and hone their skills.

Looking ahead

Integrating OTS into corporate training programs, applying virtualization, and using 3-D virtual reality are but three examples of how operator training simulators have been adapted to meet the demands of the modern workforce. We expect these synergistic approaches and other emerging technologies (e.g., mobile computing, remote operations, workflow technology, and situational awareness graphics) to result in continued advances in the world of operator training simulation, contributing to a new generation of safe, productive, and profitable plant operation.

ABOUT THE AUTHOR

Peter Richmond (peter.richmond@invensys.com) is Invensys' EYESIM/OTS product manager, supporting Invensys clients in Europe, Russia, and Africa. He holds a master of science degree in chemical engineering from the University of Manchester Institute of Science and Technology, in Manchester, U.K.

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Next-gen, new-gen, co-worker—call it what you may, a robotics revolution is rolling into manufacturing, warehousing, materials handling, and supply chains worldwide

Your new robot colleague – *coming out of its cage*

By Esben Østergaard

When asked how they envision a robot, most people either think of huge, unwieldy robots working in fenced-off areas in large factories, or they think of futuristic cyberbots mimicking human behavior.

But somewhere between these two scenarios lies an emerging reality: a new class of robots, dubbed *collaborative robots* due to their ability to work directly alongside employees with no safety caging. These kinds of co-bots are poised to bridge the gap between fully manual assembly and fully automated manufacturing lines. Nowhere is that more obvious than in the small and medium businesses sector, that up until now viewed robot automation as too costly and complex to consider.

Unlike their big brothers working behind glass at automobile plants and other big assembly lines, collaborative robots are lightweight and flexible. They can easily be moved and reprogrammed to solve new tasks, meeting the short-run production challenge faced by companies adjusting to ever more advanced processing in smaller batch sizes. The automotive sector still comprises roughly 65 percent of all robot sales

in the U.S. However, the Robotic Industries Association quotes observers who believe only 10 percent of companies that could benefit from robots have installed any so far.

Lowering the entry barrier

The reason that number is so low is primarily due to three challenges now addressed by the new collaborative robots: cost, user friendliness, and applicability. Let us start with the financial issue.

Even where workers are affordable, the next generation of complex products will require assembly adaptability, precision, and reliability that is simply beyond the skills of human workers. According to the old rule of thumb, the cost of a robot is equivalent to one worker's two-year salary. But collaborative robots are closer to one fourth of that price. Combine that with the faster turnaround time that robots bring to the workplace, and robotic technology demonstrates that the offshore exodus does not make good business sense any longer.

Instead, the new robots become a high-tech currency that is changing the wage wars into a competition over increasing product quality and quick turnaround.

A plug-and-play robot

With traditional robots, the capital costs for the robots themselves account for only 25 to 30 percent of the total system costs. The remaining costs are associated with robot programming, setup, and dedicated shielded work cells. The "out-of-box experience" with a collaborative robot is typically less than an hour. That is the time it takes to unpack the robot, mount it, and program the first simple task.

This leads us to user friendliness. Instead of requiring skilled programmers, this new class of robots comes with a tablet-size touchscreen user interface, where the user guides the robot arm by indicating movements on the screen. Or, the user can simply grab the robot arm and show it the desired path of movement. The interface is compliant with most industrial sensors and programmable logic controllers. Programming for new tasks is easy—as experienced by Danish manufacturer of hearing aids, Oticon, a company impressed by the intuitive user guidance and the precision of the new co-bots. Oticon needed a flexible robot that would be economically viable for short runs. Rapid advances in medical engineering have resulted in constantly changing production processes and a broader range of hearing-aid models that require a robot to handle smaller batch sizes.

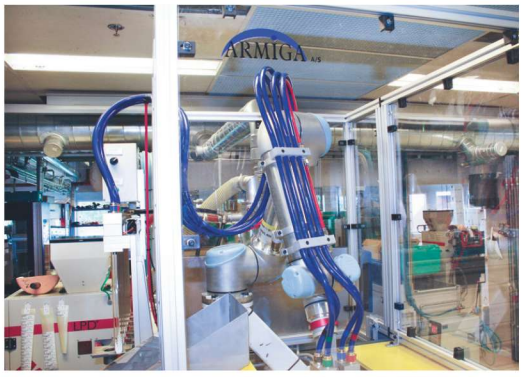
FAST FORWARD

- Collaborative robots work directly alongside people with no safety caging.
- Collaborative robots are poised to bridge the gap between fully manual assembly and fully automated manufacturing lines.
- The "out-of-box experience" with a collaborative robot is typically less than an hour, including unpacking the robot, mounting it, and programming the first simple task.

Precision handling

The new robot addresses the issues around applicability and portability not met by the traditional robots that Oticon had employed in the past. The parts for modern hearing aids are getting smaller and are often only a millimeter in size. The hearing-aid manufacturer was looking for a solution that could suction small parts out of a mold. This was impossible manually and not suitable for their "old" two- or three-axis robots that could only move laterally and vertically. If, for instance, a small part is stuck in a mold, the robot has to be able to tip it out.

It took just one day to install the robot for its new task in Oticon's molding shop. Mounted firmly to the injection molding machine, the new robot can position itself over the mold and suction the plastic elements using a specially designed vacuum system. More complex molded components are handled with pneumatic gripping tools. Because of its six axes, the new robot is very maneuverable and can rotate or tilt the parts in order to lift them quickly out of the mold. The robot works in 4–7 second cycles depending on the size of the production run and the component. Due to the optimized production process, the payback period was only 60 days.



Hearing-aid manufacturer Oticon uses a UR5 robot arm for different tasks in the foundry, where the suction tool is replaced with a pneumatic gripping tool to handle more complex cast parts. The six-axis robot works in cycles of 4–7 seconds, doing tipping and tilting moves that Oticon's traditional two- and three-axis robots were unable to perform.



At Oticon, the UR robot is securely fitted to the injection molding machine and can move over the mold and pull up the plastic items. This is done using a specially designed vacuum system that ensures the sensitive items are not damaged.





Cascina Italia automated a line packing 15,000 eggs per hour using a UR5 from Universal Robots. Employees can quickly reprogram the robot and are able to work right next to it without safety caging. The factory floor at Cascina was not laid out to accommodate separate robot automation, so a portable robot that can quickly be moved between job tasks proved crucial for the Italian egg distributor.

Working within space restraints

At Cascina Italia in Italy, a collaborative robot works on a packing line handling 15,000 eggs per hour. The robot is equipped with a pneumatic gripper and fills boxes with egg trays containing 10 eggs each. The job demands precise handling and the careful placement of nine layers of 10 eggs each in a box.

Cascina did not expect to be able to use a robot for the job, but after seeing a demo of the robot at their own factory, it was easy for the egg company to visualize the benefits. Ninety days later, the new robot was operating on the line. Weighing only 11 lb., the robot colleague can easily be moved between packing lines, which is crucial for Cascina employees who handle four different egg sizes and needed a robot that could work next to them within significant space restraints.

Safety first

Safety has been a hot-button issue and the major thrust of research and development in robotics labs for some time. With human collaboration in mind, the new generation of industrial robots has rounded joints, back drivable motors, force sensors, and lighter-weight materials.

If a robot at Cascina touches an employee, the built-in force control limits the forces at contact so the robot does not cause bodily harm, adhering to the current safety requirements on force and torque limitations. In most applications, this safety feature enables the robot to operate with no safety guards after risk assessments have been conducted.

Avoiding back-breaking movements

This is the case at Scandinavian Tobacco Company, where a collaborative robot now works directly alongside employees handling the lids for tobacco tins where tobacco is packed. The new robot spares the employees from having to make back-breaking repeated movements and freed one or two employees who previously performed the tasks by hand. They now carry out other tasks at the factory. There was no room to screen off the robot in the setup at the factory, so employing a collaborative robot simplified the setup and costs considerably.

Scandinavian Tobacco developed their own gripping tool and had one of their technicians do the initial programming. This kept the know-how in the building and ensured high productivity, while avoiding downtime in production and paying for expensive external consultants. The optimized production convinced the owner to keep production in a high-wage Scandinavian country. The return on investment (ROI) for the tobacco company's new robot was 330 days.

From 45 to 70 bottles per minute

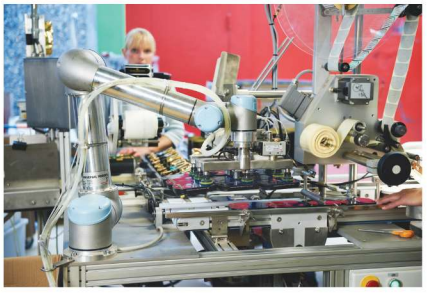
Larger manufacturers also benefit from the new robots. At Johnson & Johnson's plant in Athens, Greece, a collaborative robot has significantly optimized the packaging process of shampoos and skincare products. The robot arm works around the clock. It picks up three bottles simultaneously from the production line every 2.5 seconds, orients them, and places them in the packing machine. Manual handling processes 45 bottles per minute; robotic-assisted production handles 70 units.

The bottles are vacuum lifted and transferred cleanly without any danger of scratching or sliding. The dexterity of the robot plays a crucial role, as the label is not printed on the same side on all products and the bottles are various shapes and sizes, which means the robot has to grasp from both the right and the left. Any member of Johnson & Johnson's staff can reprogram the robot for new tasks, saving the company the cost of hiring external programmers.

A new way of approaching robotics

Above are some examples of the new generation of robots solving real-life challenges not previously addressed by robots. When it comes to human collaboration and flexible manufacturing, features of the classic industrial robot must evolve on nearly every level: from fixed installation to relocating, from periodic repeatable tasks to frequent task changes, from intermittent to constant connectivity, from no interaction



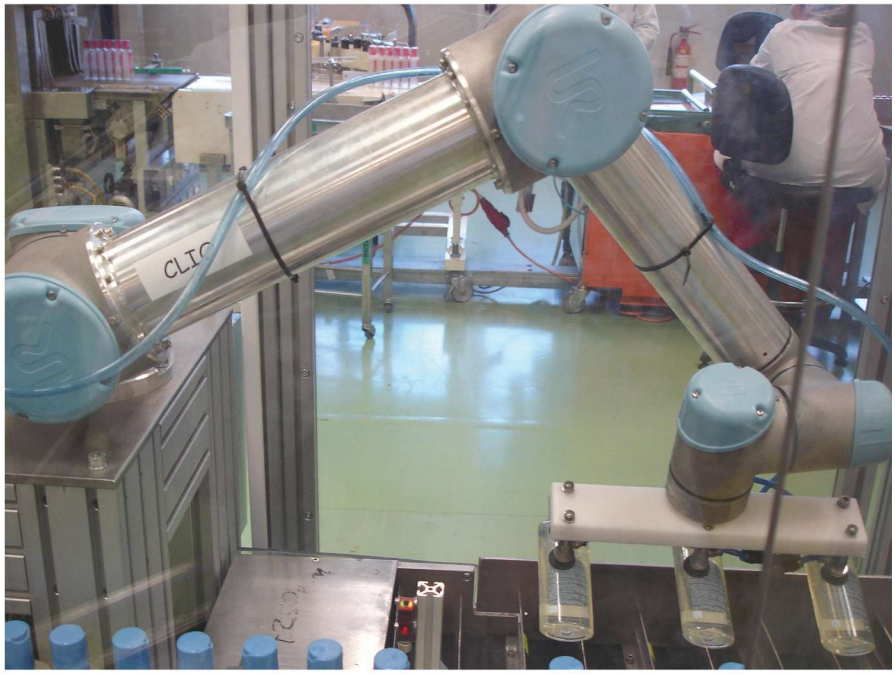


Employees are spared from repetitive back-breaking movements, packing tobacco tins into boxes at Scandinavian Tobacco, now that a UR5 handles this process. The new robot arm was well received by the employees who have moved on to less strenuous tasks.

with humans to frequent collaboration, from space separation to space sharing, from profitability within years to near-immediate ROI. The near future will see even more advances in this nascent field of robotics, changing the way we work and interact with technology.

ABOUT THE AUTHOR

Esben Østergaard, Ph.D., is chief technology officer at Universal Robots (UR), responsible for the enhancement of existing UR robots and the development of new products. During his tenure from 2001–2005 as researcher and assistant professor in robotics and user interfaces at University of Southern Denmark, he created the foundation for a reinvention of the industrial robot. In 2005, he founded Universal Robots together with two of his research colleagues. They have been granted approximately 30 patents on the technology of the robot. Østergaard also participates in national research projects and is an external examiner at several universities in Denmark.

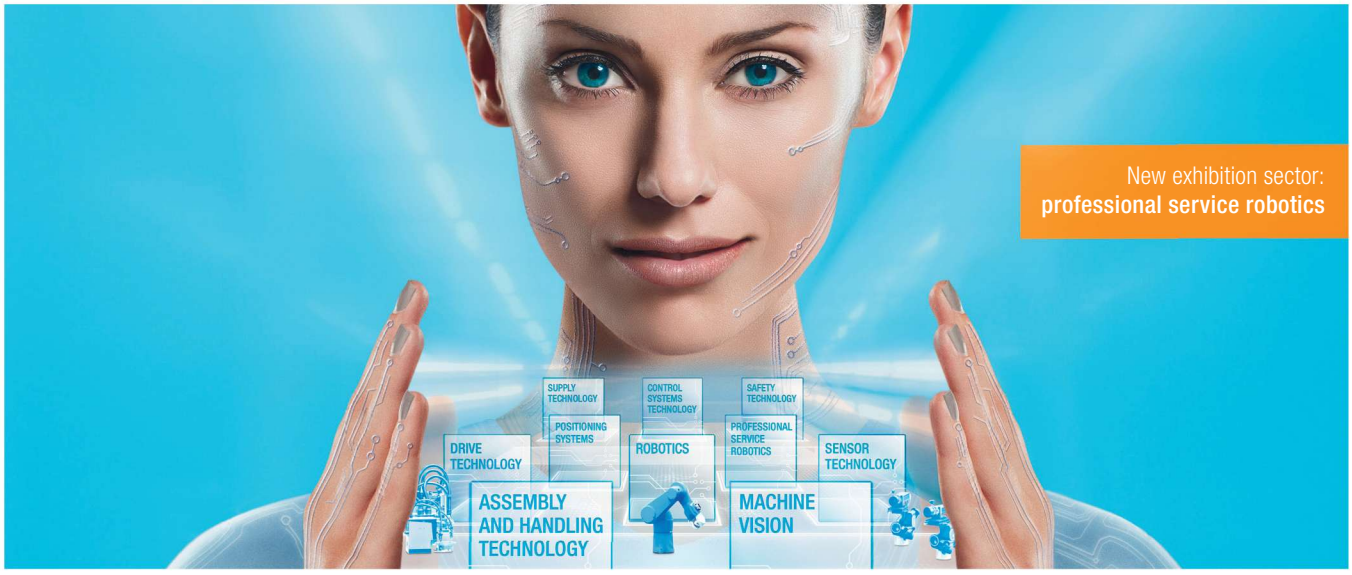


At Johnson & Johnson, the employees like working with their new collaborative robot colleague so much they gave it a name. The UR5 is now called "Clio."

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Using smart field devices to improve safety system performance

Safety monitoring software can use data from smart field devices to improve safety system performance and operation

By Guillermo Pacanins, P.E.



Figure 1. Picture of a smart instrument installed in the field.

Any process plant that handles products, feedstock, or fuels that are the least bit hazardous (flammable, toxic, or otherwise environmentally dangerous) has safety concerns. Operating in compliance with regulations and standards is a way of life for oil, gas, petrochemical, biofuel, and many commodity chemical producers. But beyond compliance, companies want and need to protect their people, equipment, and the surrounding environment.

Applicable standards include ANSI/ISA-84.00.01-2004 Parts 1–3 (IEC 61511 Mod) and IEC 61508, along with facility-recognized best procedures and practices. Compliance with these standards ensures that the plant is not simply within the letter of the law; it helps the plant operate with minimal potential for incidents and injuries.

Undertaking this effort begins with plant hazard and operability studies and the layer of protection analysis (LOPA) methodology. Some situations may call for a quantitative risk analysis, as provided by the Center for Chemical Process Safety and indicated by ANSI/ISA-84.00.01-2004 Part 3, Appendix F.

Performing a LOPA helps identify which identified hazards require safety instrumented functions (SIFs) and the required probability of failure on demand for each to lower the risk to a tolerable level. Performing a LOPA is a main step toward ensuring that requirements under ANSI/ISA-84.00.01-2004 Parts 1–3 (IEC 61511 Mod) are met.

Once the safety instrumented system (SIS) is designed and implemented according to the safety requirement specification, its operation must be maintained and monitored to ensure integrity of the SIF, and to ensure ongoing compliance with standards. Any changes to the hardware, such as new equipment, new field devices, different products, or different specified operations and processes must be taken into account using a management of change procedure. Any malfunctions or other process issues must also be accounted for, typically by proof testing and monitoring the SIS along with its associated field devices, such as sensors, instruments, valves, and logic solvers (figure 1).

Real-time safety monitoring software improves the integrity of process safety systems and ensures compliance and safe operation. Companies can enhance the results generated by the software with the information supplied by SISs, plant automation systems, and their associated smart field devices. All these systems and their associated components

must be maintained, a task that can be eased by using smart field devices.

Safety systems need maintenance too

In a process plant that runs well, the safety system can fade into the background, because it has a low daily demand rate. Nonetheless, field devices connected to an SIS still need maintenance. Many plant accidents have been caused by a neglected safety system field device not working properly when called upon in an emergency.

The reality of thinly staffed process plants is that the operations and maintenance professionals charged with this time-consuming and complex task also have to watch over the other plant assets that support regular production. They are responsible for availability, productivity, and so on. Since the SIS does not affect these areas under normal circumstances, it can become a secondary concern, or slide even further down the list of priorities.

To make matters worse, field devices that are part of the SIS do not always employ the latest technologies. They often do not have the capability to provide information to the main plant automation system, an asset management platform, a computerized maintenance management, or other related systems. There may be no alternative to sending an individual to a given field device and inspecting it where it is installed, a task that is often postponed.

All SISs depend on field devices for their information, many of which are discrete (on/off), plain 4–20 mA analog, 24 VDC, or some other analog signal type. Each device provides its primary variable and nothing more. This does not have to be the case, because smart field devices can produce extensive diagnostic and other information.

Many field sensors, instruments, and valve actuator positioners installed in the past 10 or even 15 years have some diagnostic capability built in. In other cases, dumb field devices can be upgraded to smart ones, either through retrofit or replacement. In either case, an SIS that is capable of gathering more diagnostic information from each field device greatly improves the quality of data available from these systems, and ultimately makes life easier for the process automation professionals responsible for the SIS.

However, even if all needed data is available, users must still make sense of the information. Volumes of raw diagnostic data must be transformed into useful information that guides maintenance efforts and promotes correct operation of the SIS and other related systems. This is not an easy task, as the relatively small number of plants that operate effective asset management programs indicates. Still, there is a way to improve safety system operation without unduly burdening plant personnel, and it starts with smart field devices.

FAST FORWARD

- Many process plants have smart sensors, instruments, and valves installed as part of their safety systems.
- These smart field devices can provide a host of useful information to the safety system.
- Safety monitoring software helps make sense of the information from smart field devices by turning their raw data into actionable information.

Advantages of smart field devices

When applied effectively, using diagnostics from smart field devices has a variety of benefits, which are summarized in table 1 and detailed below.

1. Diagnostics can indicate out-of-spec instrument operation. Many field devices used in an SIS are more complex than a simple level, pressure, or temperature switch. As a result, there are ways they can malfunction or drift out of the normal range. Diagnostics can indicate these safe failures where a device has malfunctioned without causing an alarm or an incident. This allows operators to compensate until the device can be repaired or replaced.
2. Diagnostics can indicate failure of communication links. A field device that is functioning properly but cannot communicate due to a network failure is still a failed device. However, with the right diagnostic information, operators can isolate the problem as a network issue more quickly and save time troubleshooting. In some cases, a workaround can be created, such as reverting to a 4–20 mA signal.
3. Diagnostics can predict incipient failure. Smart field devices have powerful capabilities

Table 1. Benefits of using smart field devices in safety systems

1. Diagnostics can indicate out-of-spec instrument operation.
2. Diagnostics can indicate failure of communication links.
3. Diagnostics can predict incipient failure.
4. Smart instruments can ease the task of redundant system design.
5. Process data, in addition to the process variable, can improve safety system performance.
6. They have the ability to automate some testing protocols, such as PST for ESD valves.
7. Safety instruments can feed information to the BPCS in appropriate situations.

to diagnose their own internal systems, to the extent that many can determine when their own circuits are beginning to show signs of degradation. This information can be sent to the asset management system, so operators have the maximum amount of time to correct the problem before an outright failure.

4. Smart field devices can ease the task of redundant system design. Many new smart field devices incorporate capabilities for what is essentially internal redundancy and diagnostics. Clever designs that include voting schemes and redundant circuitry provide the kinds of functions that otherwise would have to be built into the safety system. Moving that kind of capability into the field reduces the complexity of the centralized processing. These features include fail-safe, fail-tolerant with redundancy one out of two voting with diagnostics (1oo2D), and 2oo3D—which allow single smart devices to achieve higher safety integrity level values.
5. Process data in addition to the primary process variable can improve safety system performance. Many smart instruments are multivariable devices that make measurements in addition to the main variable. For example, a pressure instrument usually requires an internal temperature measurement to correct the pressure reading. While that temperature reading would not ever be used for a safety function, this "free" data can be helpful in troubleshooting or diagnosing process problems.
6. Smart field devices have the ability to automate some testing protocols. Partial-stroke testing (PST) of emergency shutdown (ESD) valves is now a regular practice in many plants. PST and full-stroke tests follow very strict schedules to fulfill the requirements of relevant standards. Smart valves can digitally connect their actuators and positioners to sophisticated asset management platforms that can be programmed to carry out these tests and record necessary performance data with little or no operator intervention.
7. Smart field devices can feed information to the basic process control system (BPCS) in appropriate situations. Traditionally, most field devices in safety systems were simply discrete: they went from off to on when a liquid level, pressure, or temperature rose too high or dropped too low. As those devices are replaced with smart counterparts that provide scalar digital data, the BPCS can productively use that information. As long as the basic safety function is not compromised, there is no reason that the safety level instrument in a tank cannot report its information to the BPCS, eliminating the need for another field device.

Process data in addition to the primary process variable can improve safety system performance.

Smart field devices can provide a host of useful information, but are most effective when supported by safety monitoring software.

Converting raw data into useful information

Without a plan supported by the right analytical tools, the flood of data from smart field devices can quickly overwhelm users, causing them to neglect useful information. Fortunately, safety monitoring software tools that can turn this data into actionable information are available.

A safety monitoring software platform is typically PC-based and receives information from existing control or safety systems via a digital data link (figure 2). A safety monitoring system can fulfill several critical functions (table 2).

1. It provides visualization of real-time risk exposure based on actual operating conditions. Once the SIS moves beyond simple safety switches, safety monitoring software can draw more information from the larger group of smart field devices and watch for changes in the risk landscape. Even if no devices have

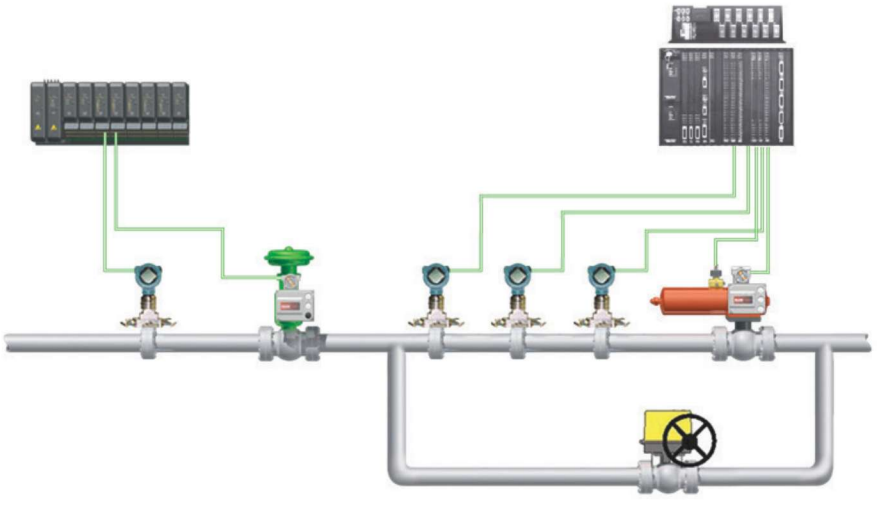


Figure 2: Diagram showing smart instruments connected to a control or safety system, which is in turn connected to a PC running SafeGuard Sentinel

Table 2. Benefits of using specialized software to monitor safety system operation

1. Provides visualization of risk based on actual operating conditions
2. Monitors changes in risk levels over time
3. Provides contingency plans to deal with safety incidents

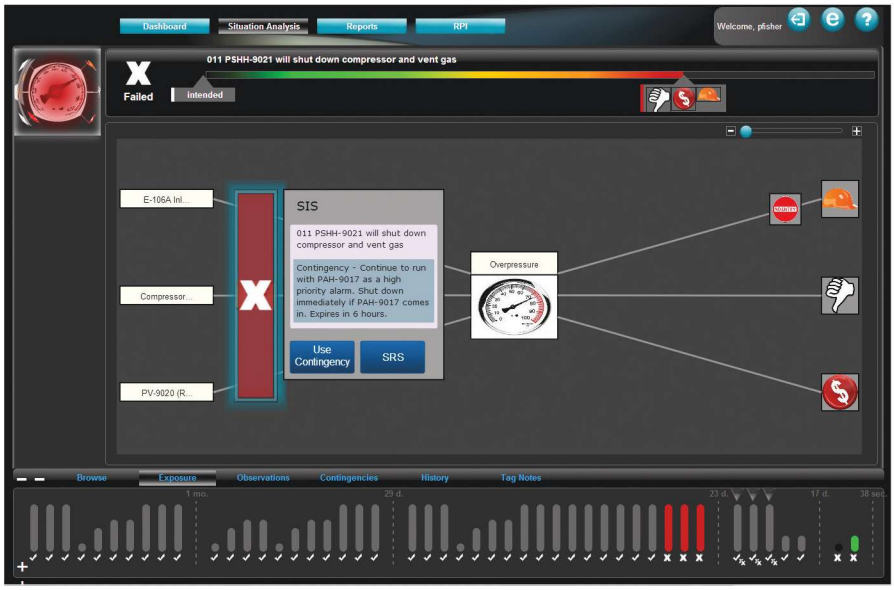


Figure 3. Screen shot from SafeGuard Sentinel

actually tripped, the system can detect a changing situation that is building toward a higher risk level.

2. It monitors changes in risk levels over time. As the safety monitoring software gathers data over time, it can determine the characteristic operating levels. If those levels begin to move or if changes in production or recipes generate new conditions, the safety monitoring software evaluates the evolving risk profile and determines if the underlying assumptions in the original design still apply. Many process manufacturing environments are dynamic due to changes in manufacturing techniques, sources of feedstock, production levels, and so forth. These variations can also change the specifications of the SIS, and safety monitoring software can guide continuous evaluation to ensure that the SIS is working as originally designed.
3. It provides contingency plans for safety incidents. Few things can make a bad situation worse than a poorly trained, overwhelmed, or panicked operator making the wrong decisions in a crisis. Many studies have shown that people are often the weakest link in a safety chain. Safety monitoring software can have embedded elements that guide operators through difficult situations, reducing the likelihood of incorrect responses. This is important, as even an experienced operator may make a wrong choice in a new situation, which is often the case with a safety-related incident.

Safety monitoring software provides the benefits outlined above by supplying information to operators in easy-to-understand formats. It turns raw data into graphs, charts, and diagrams that show overall safety system performance at a glance, while also allowing operators to drill down to reveal details (figure 3).

Although smart field devices and safety monitoring software can greatly improve safety system performance and operation, there is an art to applying these tools to optimize plant operations.

A delicate balance

Safety systems can fail in two ways: they may be unable to respond as planned during a crisis, allowing a critical situation to escalate. Or, a system may create a spurious trip and shut down the operation when no actual threat exists. Even though this is called a safe failure, it is disruptive to production and costly. There is also a temptation for operators to manually override or bypass these safe failures, which can create very dangerous situations.

An effective safety system depends on a chain of events and devices. Field devices feed data to the SIS and to the BPCS, which in turn supplies information to the safety monitoring software. But like any other systems, the devices and software tools that monitor real-time risk exposure are only as good as their users, who must possess the required level of expertise to understand the risk in the process, the SIS automation, and the safety requirements of the process.

Those responsible for maintaining the system must walk a fine line between having mechanisms that truly protect the plant, its people, and the environment—against having mechanisms that are too sensitive and trip unnecessarily. Adding a higher level of hardware sophistication can contribute to a safer plant, but at the risk of excessive complexity if intelligent design is not employed.

The best solution in many cases is to use safety monitoring software to distill the data from smart field devices and other sources into easily understood and actionable information that can improve the operation of the safety system.

ABOUT THE AUTHOR

Guillermo Pacanins, P.E., holds a B.Sc. in electrical engineering. He is a certified TÜV Rheinland functional safety expert and has more than 27 years of experience with process controls and functional safety in process industries. He serves as a system designer, workshop presenter, and trainer for ACM Facility Safety, where he holds the title of safety lifecycle leader/educator.

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New HMI alternatives improve operations and cut costs

Today's businesses demand easy and inexpensive HMI remote access along with the ability to quickly retrieve and act upon plant operating data

By Jeff Payne



Faster speeds, lower costs, and greater connectivity are essential to the success of all businesses, including manufacturing. With the introduction of lean manufacturing, Six Sigma, and other continuous improvement strategies, operations must become more flexible, easier to manage, and less expensive.

To achieve these goals, businesses demand user-friendly human-machine interface (HMI) graphics, the ability to drill down quickly for alarms, overall equipment effectiveness dashboards, and more. In addition to having reliable information at their employees' fingertips, companies also want better remote HMI access from a variety of devices. Finally, they want more intuitive interfaces to reduce training costs and improve operator performance.

Fortunately, there are HMI packages available that can fulfill all these requirements. The latest HMI software solutions shorten development time, offer better remote access, and have more intuitive interfaces to reduce the learning curve for operators and other users. Overall, these enhancements facilitate a more mobile and productive workforce while reducing training and equipment costs.

Is a PC the best choice?

When selecting an HMI, the first decision is whether or not it will be based in a PC. PC-based HMIs have the best performance, the most features, and the easiest connectivity—but also have the most expensive software licensing costs. HMIs that are not PC-based typically run on embedded operating systems. These embedded HMIs are much less expensive than PC-based HMIs in terms of software licensing costs, but are also less capable in performance and feature richness.

FAST FORWARD

- Users must first decide whether their HMI should be PC-based or embedded.
- Once this decision is made, they can determine the best methods of remote access.
- Technologies borrowed from smartphones and tablets are rapidly improving remote access.

With respect to hardware, costs depend on whether the HMI will be in the controlled environment of a control room or an office, or on the plant floor. In a controlled environment, a PC is less expensive than a similarly sized embedded HMI. But on the plant floor, an embedded HMI is much less expensive than an industrial PC. For many applications, connectivity and remote access will drive the PC versus embedded decision.

Mobility is no longer just an option

No one can now imagine a successful company that does not have remote access to business systems; the same will soon be true for automation. With just five operators being hired for every 10 that retire, mobility is essential for increasing productivity. Employees can no longer spend the entire day in the control room or one area of the plant, thus HMI remote-access capabilities are mission critical.

Mobility is often provided through wireless and cellular networks. Although security is a concern, wireless networks are rapidly becoming an accepted medium of communication in industrial environments. Lower installation and maintenance costs along with improved security have made wireless attractive to automation companies. Wi-Fi and Bluetooth are affordable, fast, and easy to use. And thanks to the evolution of encryption and the IEEE.802.11X-based Extensible Authentication Protocol, user devices must be identified to gain access to the network. Cellular networks are an attractive alternative to wireless in many cases, particularly as speeds are increasing while costs decline.

What kind of remote access is best?

When developing remote-access applications, the type of user and his or her requirements must be considered, along with costs. Evaluating these and other factors will determine the appropriate method of access from the listed options.

Note that most companies will use multiple methods of remote access, mixing and matching depending on the specific needs of the particular user, on the costs involved, and on the technical capabilities of the organization.

Most every PC-based HMI will support all four remote-access options, whereas most embedded HMIs will only support web browser and app access. Embedded HMIs are meant to stand alone for the most part, whereas PC-based HMIs are often part of a networked system of PCs and thin clients.

Remote access via a PC has the best remote user experience. PC remote access is very low in hardware cost if the user already has a PC and is using it for other purposes. But networking a remote PC can be expensive, and software licensing costs are high. If the remote-access

Types of remote-access options
PC
Thin client
Web browser
App

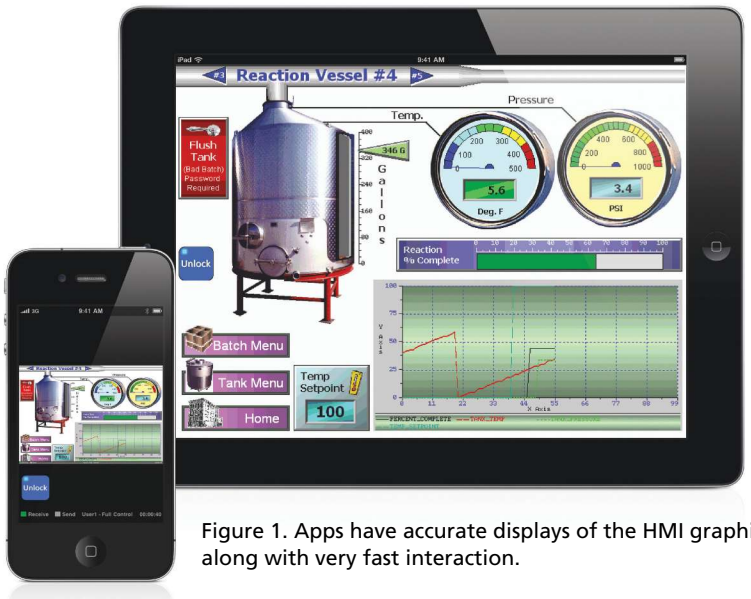


Figure 1. Apps have accurate displays of the HMI graphics, along with very fast interaction.



Figure 2. HTML5 support offers HMI app access from a wide variety of portable devices, without the need for costly and time-consuming custom software development.

PC needs to be mounted on the plant floor, costs will be very high.

A better alternative for plant floor remote access is a thin client, which has nearly all the capabilities of a PC at a much lower cost, both in terms of software licensing and especially hardware. Not only is a thin client less expensive to purchase, it is also much less expensive to mount on the plant floor. It will have much less processing power than a PC, and hence generate less heat. Thin clients are also much less expensive to maintain than PCs, as most all software is installed and maintained at the PC-based server as opposed to on the thin client.

Whether the HMI is PC-based or embedded, browser and app access are two other remote-access options that should be considered and evaluated.

Browser or app?

In today's plants, operators must monitor and control more processes in multiple places. They must be able to quickly and easily access HMI information from mobile devices from any area in the plant or outside it.

Moreover, the growing trend of "bring your own device" (BYOD) means companies can save money on hardware and software. However, BYOD also means authorized users will be accessing HMI data from a wide array of devices such as iPhones, Androids, iPads, tablets, and PCs.

In response to their customers' needs, HMI suppliers have designed products for remote interaction via a web browser or an app. Both options have advantages and disadvantages, depending on how they are accessed.

Browser access typically works seamlessly for PC users, but the web server screens do not often scale well to the smaller displays on handheld devices. The screens can take too long to load, and only a small portion of the display may be visible. Browser-based access is often free in terms of software licensing costs, but the expense of designing, deploying, and maintaining the network and web servers can be substantial.

Apps are usually the faster, easier choice for handheld device users. Free or very low-cost apps are created for smaller screens to improve download times, and software licensing costs for host software are usually either zero or very low. Apps provide users with better visualization and much faster interaction, two key attributes for remote access (figure 1).

The downside to using apps, however, is they are usually first developed for iPhones and iPads, with apps for other devices, such as the Android operating system (OS), lagging. This is because a large number of vendors sell tablets and smartphones, and there is consequently a lack of standardization.

Android-based devices alone come in at least seven screen sizes. The lack of standardization has made it cost prohibitive for HMI suppliers to develop an app for every handheld device's screen size and OS. Moreover, users do not want to wait months or years for the app to be developed for their particular device. Fortunately, there is a solution, and it involves the adoption of the HTML5 standard.

In today's plants, operators must monitor and control more processes in multiple places.

Software standards to the rescue

For thousands of users, it seems as if the ability to use an Android or tablet for HMI access is merely a dream. However, the releases of Windows 7 and 8, which both have HTML5 support, promise users fast remote connectivity, regardless of their device type. HMI software packages with HTML5 support enable users to get simultaneous remote access from almost any device, without waiting for an app to be created for their specific device. The "design once and deploy everywhere" approach allows delivery of remote-access capability by the software supplier to any device with HTML5 support, regardless of the OS or screen size (figure 2).

App access was initially a feature restricted to Apple platforms, and that is still the case for some HMI software suppliers. But the demand for apps that work with smartphones and tablets from other vendors is so great that HMI suppliers will start to develop apps that use HTML5.

Multitouch improves performance

It is easy to see how reliable and fast remote access benefits businesses, but

multitouch capabilities may seem like more of a gimmick to entice younger workers. While it is true that tomorrow's employees will likely have rarely touched a mouse or other pointing device, multitouch enables all workers to improve performance.

Multitouch HMI offers many advantages over single-touch screens, keyboards, and pointing devices. It recognizes the position of several touches and finger movements, which are referred to as "gestures." Training time is shorter, because these gestures—pinch, zoom, swipe, and more—are the same as those used for smartphones and tablets.

By using gestures, operators can execute commands up to three times faster than those performed on regular touch screens. The ability to easily manipulate objects on the screen also helps them find the exact location of a potential problem quickly. For example, an operator might get an alarm about a certain piece of equipment. Using multitouch technology, he or she can easily rotate a piece of equipment on the screen, zoom into a specific area, and then magnify the area—all without once lifting a finger from the screen. When an event occurs, an operator can quickly zero in on areas of interest, instead of wasting time using drop-down menus or scrolling through multiple screens.

The fewer moving parts of multitouch tablets make them a better choice for workers who visit dusty, wet, and corrosive environments. Industrial tablets have the durability required for these areas, and many can be operated while wearing gloves. Furthermore, they can improve worker safety through the creation of commands that cannot be performed unless both hands are on the screen.

Although it is unlikely that businesses are going to swap their functioning screens for new multitouch ones, it is highly probable they will replace worn-out screens with multitouch capability as the price for these devices drops. Multitouch functionality is also expected to become more ubiquitous due to the integrated support for the technology in new Windows operating systems. Eventually, all screens will likely have multitouch capability, so it is smart to select an HMI package that supports it.

While the automation world can be slow to implement change, it is being forced to increase performance and decrease costs and will need the technologies that can help accomplish these goals. Forward-thinking HMI vendors are ready for these challenges. These suppliers see BYOD as being the norm rather than the exception. They are offering solutions with HTML5 support, so companies can quickly and easily connect remotely to a wider array of portable devices. They comprehend that multitouch is new to automation companies, but they realize the sea change that is coming when these enterprises discover how much they can improve operations.

More than ever before, seemingly future technologies and applications are rapidly becoming the staples of today's businesses, and the companies that implement these advancements will stay ahead of the competition.

ABOUT THE AUTHOR

Jeff Payne (jpayne@automationdirect.com) has worked in industrial automation for more than 25 years; he is the product manager of automation controllers and interface software at AutomationDirect.

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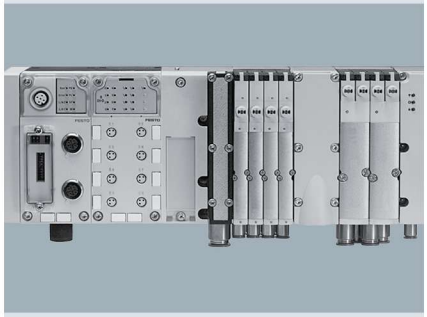
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Growing future automation professionals

FIRST inspires and educates youth to become automation professionals

By Bill Lydon

Dean Kamen, the founder of FIRST (For Inspiration and Recognition of Science and Technology) and CEO of DEKA Research & Development, has created an organization to energize young people to become science and technology heroes. The mission of FIRST is to inspire young people to be science and technology leaders, by engaging them in exciting mentor-based programs that build science, engineering, and technology skills, that inspire innovation, and that foster well-rounded life capabilities, including self-confidence, communication, and leadership. The FIRST Robotics and other programs are designed to accomplish this mission. The Automation Federation (AF) and ISA are supporting FIRST as strategic alliance partners.

FIRST's programs give young people knowledge and experience to make informed decisions about pursuing opportunities in science, technology, and engineering. With support from three out of every five Fortune 500 companies and $14 million in college scholarships, the not-for-profit organization hosts the FIRST Robotics Competition (FRC) and FIRST Tech Challenge for high-school students, FIRST LEGO League for nine- to 14-year-olds (nine- to 16-year-olds outside the U.S. and Canada), and Junior FIRST LEGO League for six- to nine-year-olds.

Why should you care?

Using technology to advance the economies of the world will require a pipeline of knowledgeable, motivated, and enthusiastic talent. Dr. Peter Martin, Invensys vice president, gave a passionate keynote at the ISA Automation Week 2013

conference, "The Future of Automation," describing how automation can solve the world's biggest challenges, including efficient energy, safe water, food production, health improvement, and a clean environment. For example, Martin noted over half a million babies a year die because of tainted water. That is inexcusable, since this can be changed with the application of technology. We need to motivate more young people to join the automation profession.

FAST FORWARD

- FIRST has innovative programs for young people to build self-confidence, learn, and acquire life skills that motivate them to pursue opportunities in science, technology, and engineering.
- A young ISA member reflects on how her FIRST experience led to a career in the automation profession.
- There are great opportunities to contribute time and talents to make a difference in young people's lives.

It works!

Danaca Jordan, staff engineer at a major chemical company and attendee of ISA Automation Week 2013, became interested in automation due to her FIRST Robotics experience. She related the excitement of those events. In preparation for each tournament, members of the CRyptonite team would dye their hair bright green. Parents, friends, and fans (i.e., siblings) would wave signs, wear green shirts, and throw beads while they cheered them on. She wore a nylon fluorescent green cape with the CRyptonite logo her FIRST team created. Also, the team had the disc jockey at the event blast the song "Kryptonite" by the music group 3 Doors Down every time their robot took the field. As a senior, she and the co-captain of the team would talk to the technology judges, selling them on the team's new designs while the driving team did last-minute maintenance. She noted that it has been a decade since she was a member of FIRST Robotics Team #624 from Cinco Ranch High School, but she still remembers the competitions and six-week robotic build phases as the most exciting part of high school. Jordan said this program was the reason she went into technology and earned a Bachelor of Science in chemical engineering at the University of Houston.

Reflecting on this experience, she really appreciates the entire crew of adults supporting their team that made the experience possible, including the teachers who helped them set up a special robotics study hall. They could build robots and write control programs on school time and equipment. One of their sponsors, Oceaneering, let them use its machine shop after hours. Oceaneering's staff would stay late in the evening to make sure they used the tools safely. She proudly exclaims, "I was one of the few 16-year old girls I knew who could both use a drill press and solder wiring. Other mentors from BP and various engineering firms would participate in marathon design sessions, each one trying to guide us away from the impossible without taking over our design." The team's parents eventually formed a booster club to support community events and help raise funding for ever-evolving designs. "Everyone made time to help us, and I didn't realize then how precious spare time is." She is thankful to all the people who helped.

After high school, one of the team's mentors helped secure her first engineering internship that helped her pay for college and launch her career. Jordan acknowledges she uses the skills learned in robotics daily for both technical efforts and project management in her role as a staff engineer. "Most importantly, robotics has provided the challenge and camaraderie to keep engineering and automation exciting for me," she explained. "It is my turn to support the upcoming engineers in training, and I can't wait."

Danaca Jordan is an active, contributing ISA member. She urges ISA members to find some way to share their experience with the future engineers and programmers who are in school now.

Strategic alliance

AF and ISA have formed a strategic alliance with FIRST. The purpose of this alliance is to build the next generation of automation professionals by promoting the importance of science, technology, engineering, and math (STEM) in K–12 education. Through this alliance, FIRST, AF, and ISA will mobilize their joint resources to collaborate and promote K–12 STEM education through after-school participation in the four FIRST robotics programs.

ISA section members around the world are invited to contribute at the grass-roots level. Kamen stated the need for mentoring, "The importance of pairing our students with an engaged and accomplished group of engineering and automation professionals cannot be under-



estimated. Our students will get the opportunity to work with professionals whose work reflects the typical challenges and creative rigor that is at the heart of our program."

FIRST will assist the AF and ISA members and affiliates across the country with resources, such as instructional materials, guidelines for starting robotics teams, and marketing support. AF and ISA staff, members, and volunteers will be matched to FIRST affiliate and operational partners and will support events in their geographic areas to collectively build local collaboration. With help from FIRST, an earned recognition program will be established for AF community members participating in FIRST robotics.

Get involved

FIRST is about practical, experiential learning, and experienced ISA members can be valuable contributors, making a difference in the future careers of students. There are a number of ways ISA members can get involved in FIRST:

- Volunteer at an event. Each competition depends on an abundance of volunteers with a broad spectrum of talents to support operating needs and competition demands.
- Join an existing team. If you are interested in becoming a FIRST mentor, visit www.usfirst.org/whats-going-on to find events and teams in your area. You can attend a regional competition in your area to check it out. Attendance is free.
- Connect. You can introduce yourself to the FIRST regional director near you (www.usfirst.org/regional-contacts).

The FRC is a major event that is a unique "varsity sport of the mind" that helps high-school-aged young people discover how interesting and rewarding the life of engineers and researchers can be. The FRC challenges teams of young people and their mentors to solve a common problem in six weeks using a standard "kit of parts" and a common set of rules. Teams build robots from the parts and enter them in competitions designed by Kamen, Dr. Woodie Flowers, and a committee of engineers and other professionals. FIRST redefines winning for these students, because they are rewarded for excellence in design, demonstrated team spirit, gracious professionalism, maturity, and the ability to overcome obstacles. Scoring the most points is a secondary goal. Winning means building partnerships that last. Visit www.usfirst.org for more information.



ABOUT THE AUTHOR

Bill Lydon is chief editor of *InTech* magazine. Lydon has been active in manufacturing automation for more than 25 years. He started his career as a designer of computer-based machine tool controls; in other positions, he applied programmable logic controllers and process control technology. In addition to experience at various large companies, he cofounded and was president of a venture-capital-funded industrial automation software company. Lydon believes the success factors in manufacturing are changing, making it imperative to apply automation as a strategic tool to compete.

View the online version at www.isa.org/intech/20140206.

Eliminating black boxes in safety applications

By Ken O'Malley, P.E., CFSE



In the past, many safety controllers were typically procured and supplied as mysterious black boxes with few or no links to other safety and automation systems, the antithesis of an open system. Instead of a black box, process automation professionals can design their own safety system that integrates with other automation systems and has superior performance, more uptime, and faster diagnostics.

Black boxes are being replaced by open controllers in burner management systems (BMSs). That solution is a safety instrumented system (SIS) built around a programmable safety controller that will conform to all codes and add a host of useful features and connectivity options. Today's marketplace offers many safety products that are competitively priced and relatively easy to use.

Standards permit users to consider a BMS to be an SIS with its hazardous operations; layer of protection analysis (LOPA); safety requirements, design, configuration, factory, and site acceptance testing; ongoing management; and periodic testing adhering to the complete safety life cycle as mandated by the ANSI/ISA-84.00.01-2004 standard.

With a traditional BMS controller, process switches that detect dangerous process deviations are wired to nonprogrammable, purpose-built black box logic solvers. Upon deviations, the controller isolates fuel sources to the combustion chamber as required, but gives little or no feedback to the operator. To alleviate this issue, smart transmitters with internal diagnostic capabilities can be used in lieu of switches, along with safety-rated logic solvers based on programmable logic controllers (PLCs). This solution provides greater process awareness for operations and easier troubleshooting for maintenance following an equipment trip.

One of our customers recently spent two days determining the cause of a BMS black box controller trip. The facility had a backup boiler, so it avoided costly extended downtime, but troubleshooting costs were high. This customer is now working with us to replace its black box BMS logic solver with a PLC-based BMS.

In most cases, downtime is very costly, so there is a great need to use an ISA-84 performance-based approach that gives designers the flexibility to add redundancy and trip voting to reduce nuisance trips due to single-point failures. This approach typically results in at least one safety integrity level (SIL) 2 safety function.

Achieving SIL 2 with a general-purpose PLC places a heavy burden on the end user to demonstrate that the PLC is suitable for an SIS application through a proven in-use analysis. Additionally, using a general-purpose PLC in a BMS application requires the addition of an external watch dog timer (WDT) to protect against undetected failures of the PLC. If a safety PLC is used, the WDTs may be optionally omitted if approved by the authority having jurisdiction, eliminating nuisance trips associated with the failure of these devices.

A few years ago, I was participating in a LOPA session at a customer's facility where the site's lead safety engineer interrupted the meeting and handed a WDT to me. After an all-night investigation following a boiler trip, she had isolated the cause to the faulty WDT. Our company had installed the BMS on their boiler a year earlier. Even though an SIS BMS with a safety controller was used and WDTs were therefore not required, the customer's project team had elected to keep the WDTs. But by reducing nuisance trips through the elimination of single points of failures like a WDT, the system becomes safer, because many incidents occur while placing equipment back into service following a trip.

It is important to note that even when the BMS is designed through an ISA-84-based SIS approach, the end user is still responsible for compliance, not the BMS vendor or the system integrator. And in order to meet the equivalency clause of the code of record, the complete life cycle as outlined in ISA-84 must be followed, including SIL selection, verification, and ongoing functional testing at the calculated test interval.

A BMS can be implemented in strict accordance with the appropriate code of record through a prescriptive-based approach using a black box controller. But, implementing a BMS as an SIS with a programmable safety controller offers advantages including reduced nuisance trip rates, improved troubleshooting information, and greater process and system health information.

ABOUT THE AUTHOR

Ken O'Malley, P.E., CFSE, is a degreed electrical engineer with 25 years of automation engineering experience. He is currently executive VP engineering technology at aeSolutions (www.aesolns.com). O'Malley has participated in the development of numerous PLC-based products for fire and gas and burner management system applications, and several of these products are now available with FM approval.

ISA well-situated for the challenges and opportunities of 2014 and beyond

By Peggie W. Koon, Ph.D., ISA President 2014



First and foremost, I am extremely excited to serve as 2014 President of ISA, an organization that has contributed so much to both my professional and personal growth. You see, my presidency is a celebration of two decades of ISA membership and service. Twenty years ago, after accepting an invitation to speak at an ISA technical conference, I decided to join an ISA technical division. Since then, I have been engaged with the Society in a marvelous exchange—of ideas, service, knowledge, and friendship. Like so many ISA leaders, I have benefited from ISA membership. I have been able to participate in and give time and effort to ISA—a Society that has reciprocated by providing me amazing opportunities to learn and lead. I have also enjoyed access to outstanding technical resources and have been blessed to work with and benefit from so many talented professionals.

Challenges and opportunities

Exciting, challenging, opportunity—these are the three words that first come to mind when I think of ISA and the year ahead. 2014 promises to be a year of significant change for our Society, not just because of new leadership and governance, but because of the ever-changing world in which we must operate.

For example, our understanding of the global automation community is changing. As we begin to look for new growth opportunities, our view must expand to include the various industry segments and markets that depend on automation every day. With this new perspective comes the recognition that ISA's ability to provide products and services for automation (professionals and industries) extends far beyond the process industries—a market in which we have thrived for 67 years. We also enter 2014 with emerging technologies that have created new opportunities for automation around the world, and have changed the roles, responsibilities, and needs of automation professionals. All of these developments affect ISA, its spectrum of products and services, and its *audience*.

ISA's success both now and in the future depends on its ability to seize these opportunities while remaining relevant to its *audience*—to automation professionals and to the industries and entities they serve. *And how do we remain relevant?* We must continue to deliver value—both to individual members and to the global automation community—and we must do this with *excellence*.

In response to these challenges, ISA has begun to develop a strategic road map that will clearly define our mission, vision, and goals in 2014, and the products and services to be delivered, the partnerships to be secured, and marketplace opportunities to be explored, and more. We have already begun having discussions about the new road map with ISA leaders and will continue to engage members, volunteers, and staff in the ongoing conversations in the weeks and months ahead.

Priorities for 2014 and beyond

In this new year and beyond, ISA must look both inward and outward to secure new growth opportunities, to increase awareness of its value proposition, to strengthen its brand, to tap into new revenue streams, to boost membership, and to develop a proactive plan for engaging the next generation of leaders. To do so, ISA must be successful in three vital areas: operations, collaboration, and innovation.

We need to closely examine the operations at all levels of the Society to optimize processes and resources. Secondly, we need to seek out opportunities to collaborate, both within the Society (across geographic, technical, and operational boundaries) and with external entities (including government, academia, Automation Federation sister organizations, other professional organizations and communities, and members of the automation industry) where synergy either already exists or can be created. Strategic partnerships are key to collaboration and operational excellence.

And finally, we must develop innovative solutions and approaches across the organization. Harnessing the right technologies and resources will be essential. By doing so, we will not only enhance existing products and services, but we will develop new products, services, and partnerships that will position us for the future.

Realizing our potential requires a paradigm shift—a laser focus on our achievable goals, metrics that track our progress at all times, the ability to act quickly, and ongoing commitment. In 2014, I look forward to working with all facets of ISA to move the Society forward in these areas.

A bright future for ISA

The future for ISA is extremely bright, especially since we are poised for change. Our focus on new strategic processes will help us successfully navigate through and capitalize on these changes, and assess how we "fit" so that as the markets we serve evolve, we are agile and able to respond quickly to take advantage of opportunities.

By now you are probably asking, "What about me?" Every ISA member and volunteer can help ISA grow and evolve. You may say, "I am just one person. What could I possibly do to make a difference?" One of my favorite quotes is by Edward Everett Hale: "I am only one, but I am one. I can't do everything, but I can do something."

Not only do we need your help, but you *can* help. In fact, you are already contributing by virtue of your ISA membership. In 2014, I challenge you to do even more. Seek out ways to use your expertise to serve. Introduce your colleagues and company to ISA. Make them aware of what ISA offers. Get more involved in your ISA section or division. Join an ISA LinkedIn group. Reach out to your local, national, or global ISA leadership to inquire how you can help. *Get involved!*

Together, we will make 2014 the best year ever to be a part of ISA!

ISA Certified Automation Professional (CAP) program

Certified Automation Professionals (CAPs) are responsible for the direction, design, and deployment of systems and equipment for manufacturing and control systems.

CAP question

Which of the following is true of operating instructions?

A. They require a set of books covering the total operation.
B. OSHA requires operating procedures for all installations.
C. They may be included in a functional specification or operating description.
D. ISA standards guide in the development of operating instructions.

CAP answer

The correct answer is **C**, "They may be included in a functional specification or operating description." Operating instructions may be included in a functional specification or operating description and may range from a few pages describing how to operate one part of a plant to a complete set of directions covering all parts of a facility.

Answers B and D are not correct. There are no ISA standards available to aid in developing operating instructions. OSHA requires operating procedures for all installations handling hazardous chemicals, but not for those that do not.

Answer A is not correct. Operating instructions can be printed (e.g., books or manuals) or electronic (e.g., PDF) and can cover from one step of the process to an entire operation. For each use, operating procedures are usually limited to a single operation or set of operations.

Reference: Trevathan, Vernon L., *A Guide to the Automation Body of Knowledge, Second Edition,* ISA, 2006.

ISA Certified Control Systems Technician (CCST) program

Certified Control System Technicians (CCSTs) calibrate, document, troubleshoot, and repair/replace instrumentation for systems that measure and control level, temperature, pressure, flow, and other process variables.

CCST question

A cold junction compensation is required for which type of temperature measurement device?

A. resistance temperature detector
B. thermocouple
C. infrared pyrometer
D. bi-metal thermometer

CCST answer

The correct answer is **B**, thermocouple. A thermocouple is comprised of two wires of dissimilar metals, connected at one end, which is placed in the process to be measured. This is called the *measuring junction.*

Thermocouples generate millivolt signals, which can be correlated to the measured temperature. As with all voltage signals, the millivolt signal developed by the thermocouple must be referenced to a known voltage. For example, in current loops, we often use earth ground as a voltage reference (0V).

Any connection a thermocouple makes (including the connections of the thermocouple wires to a terminal block or PLC input card) will form an additional measuring junction. Therefore, a reference junction, called a *cold junction,* is used to provide a junction with a known, stable output. This allows the thermocouple to generate a millivolt signal along the wires that can be correlated directly to the temperature reading of the measured process.

The other three temperature measurement devices listed in the problem statement do not require a cold junction reference.

Reference: Goettsche, L.D. (Editor), *Maintenance of Instruments and Systems, Second Edition,* ISA, 2005.



Hybrid temperature controllers offer more versatility

Temperature controllers have taken on PLC and HMI capabilities

By Clayton Wilson

Many process plant units have temperature loops that must be monitored and controlled. If these units are self-contained to some degree, as with a process skid or a remote unit, they often have just a few discrete and analog I/O points that must be monitored and controlled in addition to temperature. These units also often need some limited operator interface.

In the past, these remote units were often controlled by basic temperature controllers, a programmable logic controller (PLC), and a human-machine interface (HMI) terminal. But now, a single hybrid temperature controller can perform all these functions by controlling and monitoring multiple temperature loops, by acting as a mini-PLC, and by providing limited but often sufficient operator-interface functionality.

These hybrid temperature controllers excel at loop control, often employing self-tuning and basic artificial intelligence algorithms. This makes them suitable for the most demanding temperature control applications.

Feature	Traditional temperature controller	Hybrid temperature controller	Mini-PLC	Traditional large-scale PLC
I/O count	<10	<20	<40	100+
Programming complexity	Low	Medium	High	Very high
Program execution speed	100+ ms	50–200 ms	20–100 ms	10–50 ms
Connectivity options	Serial	Serial, Ethernet	Serial, Ethernet	Serial, Ethernet
Internal HMI capability	Minimal	Medium	Medium	Minimal
Loop processing capability	1	1 to 3	1 to 5	10+

Table 1. Comparative functionality of various controllers

Two forces that continue to drive the development of hybrid temperature controllers are the consolidation of instrumentation, and the migration from discrete controls to PLCs (table 1). User demands typically guide changes in products as they try to force as much capability as possible into a controller in an effort to squeeze every dollar out of costs.

In many applications, temperature control, logic control, and operator interface are all required at the process unit. In cases where a lot of computation or loops are required, a full-blown PLC-based system with a sophisticated separate HMI system makes sense. But in situations with just a few control loops and 10 to 20 other I/O points, this type of solution is cost prohibitive. In the past, many users would turn to one or more temperature controllers, a mini-PLC, and an operator-interface panel to get the needed functionality, but there is a better alternative.

To help bring costs down, some instrument manufacturers are combining temperature control, PLC logic control, and operator-interface functionality into a single hybrid temperature controller package (figure 1). This approach makes sense for a couple of reasons. First, powerful semiconductors allow complex operations to be performed at higher speeds than traditional controllers. Second, the existing operator interface on the temperature controller can also provide an interface to the logic control, and provide additional operator interface as required.

There are many benefits of combining these functions into a single instrument. Besides the reduced cost of the unit itself, there are other cost savings that emerge in the larger implementation.

- The size of the enclosure can be reduced, because less space is needed for housing one control device versus three.

- The amount of wiring and the wiring time can be minimized, as wiring does not need to be split between two controllers and the operator-interface terminal in the panel.
- Many wired connections that are typically required are no longer needed.
- Programming time and interface complexity can be reduced or eliminated.
- Operators need to learn and maintain only one PC-based programming software package, as opposed to three.

Many of these functions, such as timers, counters, and relay latches, are intuitively tied together inside the controller. This simplifies programming and speeds implementation.

Reduced need for complex HMIs

Organizations reduce costs, because they may no longer require a separate HMI on some applications, or may employ a smaller, cheaper one. Some controller features that merge well with internal ladder logic functions can work with programmable function keys, customizable displays with scrolling messages, and data entry capability. Function keys on the front of the controller can replace panel push buttons. These can be used as digital inputs to a ladder program, allowing operator interaction without the need for external hardware (figure 2).

A process alarm or a door switch can trigger the display to scroll a text message such as "Low Flow Rate" or "Door Open." This tight coupling of proportional, integral, derivative control and a ladder logic program gives more benefit at a lower cost than using a separate temperature controller with a PLC, especially where an HMI is needed.

The displays on temperature controllers typically provide custom parameters. These parameters are twofold: data display and data entry, similar to a traditional HMI. Program variables, such as values for timers or ratio factors, can be entered for use by the ladder program. Custom calculated values from the ladder program, such as BTU or flow totals, can be displayed for the operator. This type of information typically is entered via a costly HMI, but these controller hybrids can do an adequate job of replacing touch panels in cases where costs are an issue. The capabilities of hybrid temperature controllers can open up other applications, such as replacing remote distributed control system (DCS) or PLC I/O.

Avoiding costly remote I/O

A hybrid temperature controller can often be used as remote I/O for a larger PLC or DCS that is managing the entire process. A smaller, task-specific controller can be placed near the part of the process it is controlling, providing local operator interface while being monitored and directed from the host. Organizations can achieve significant savings by avoiding the need to buy I/O hardware manufactured by the PLC/DCS provider, which is typically quite expensive. In addition, the local hybrid temperature controller can offload processing tasks from the host control system, and can provide an additional level of safety by allowing remote process units to shut down safely if the host fails.

This approach is extremely reliable for collecting process information and controlling remote I/O. Many PLC-based communication protocols are available in these hybrid controllers for this type of situation: Modbus TCP/IP, EtherNet/IP, CC-Link, Profinet, and many other protocols.

The selection of a protocol for your situation will be based on many factors, including the hardware you have selected, the types of data and diagnostics you require, and the technical expertise available to engineer and implement a communications system.

Hybrid temperature controllers have certainly added significant functionality over the past few years, but there are limitations to their use.

I/O count: Hybrid temperature controllers are typically limited to about 20 discrete points and a handful of analog points. This is fine for small process units where a micro- or mini-PLC is needed, but replacing a large-scale PLC is impractical.

Execution speed of the ladder program: Logic execution in hybrid temperature controllers is



Figure 2. With all the display and programming options built into a hybrid controller, there is little need for a separate operator-interface panel, saving space and cost.

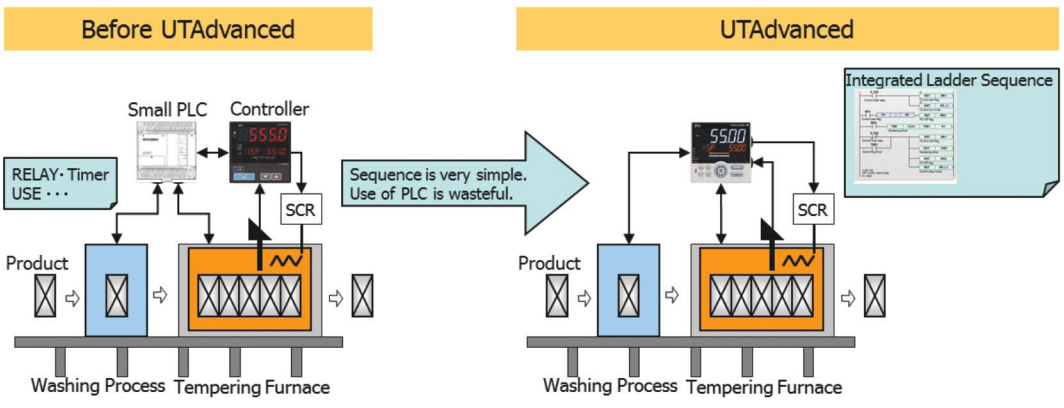


Figure 1. This function-block diagram is typical of current offerings among hybrid temperature controllers.

going to be limited to the scan rate of the controller, which is typically 50–200 ms, so high-speed switching or sequence-of-events applications are out of the question.

Programming capacity: All functions need to work within about a 1,000 ladder step limit. This is usually plenty of space given the types of programs that work with small I/O counts, but applications that must crunch a lot of numbers or perform floating-point math will chew through ladder steps pretty quickly, so caution is in order.

That said, building a group of totalizers or calculating temperature/pressure compensation for process variables does not require much in the way of ladder programming, particularly because hybrid temperature controllers are optimized for these types of functions. This leaves a wide range of applications open to hybrid temperature controllers, including the one described below.

A good application example for a hybrid temperature controller is a batch furnace where the operation calls for a guaranteed soak time for a load, but where the complexity of a profile controller is not desirable. The controller would employ a short ladder program that compares the process temperature to the set point. When the temperature reaches its set point, a timer with a user-defined time variable would start. When the time expires, the controller mode would be changed to stop, and a digital output would close to signal that the process was finished.

If a flame detector is needed for a gas-fired furnace, it could be brought into one of the I/O points. When a flame out is detected, the controller would go to manual mode and turn the output off. Another input would cause the output to go to 100 percent for a purge.

Temperature controller capabilities have increased substantially over the past few years. Some of the products that users have become accustomed to have been morphed by market forces into some truly interesting variants, such as hybrid temperature controllers. These controllers are carving out a niche with significant benefits for applications needing good temperature control capability, modest I/O logic control, decent computational capability, and limited operator interface.

ABOUT THE AUTHOR

Clayton Wilson (clayton.wilson@us.yokogawa.com) is a control instruments division manager at Yokogawa Corporation of America, Newnan, Ga.

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Mission-critical operations

By Mitchell Sepaugh

Technology advances have allowed us to expand our horizons and create complex systems to automate the supporting fabric and functions of our daily lives. These are systems that we have come to depend on and, in some cases, take for granted. We reach for the light switch without thinking that the light might not come on. We pick up the phone and rarely ever wait for the dial tone, because it is "always" there.

For centuries, we built our businesses and industries around dependable and reliable people. However, as technology becomes more dependable and reliable, we have started to shift the mundane, tedious, and repetitive tasks from people to technology. After all, technology can do things faster, more accurately and, in many cases, much less expensively than people can. Nevertheless, with all this dependable and reliable technology around us, what happens when it suddenly, picosecond suddenly, becomes unusable, broken, or—even worse—hacked? Now that core piece of technology your company relied on is offline or renegade. With the change in operating status, you start losing money or lives within the next picosecond. What do you do?

Many industries have adapted to the rise in complexity of these systems, which are comprised of different technologies, with the advent of the systems integrator and a systems engineering approach. These integrators and engineers do a wonderful job taking the requirements of a given project and pulling the right mix of technologies from their vast resources to solve the puzzle of the project in the most optimal manner. This interdisciplinary approach yields a system that, when properly integrated, creates a synergy between elements that are derived from vastly different types of technology. Depending upon the solution, the system could have electrical, mechanical, and chemical components networked via a variety of transducers with one or more electronic central processing units for both automatic control and interface to human operators. To produce such a system, the systems integrator will employ specialists in each of the required engineering or technology fields (e.g., electrical, mechanical, and chemical as needed). However, once the system is designed and installed, the customer is left to the mercy of the nines of uptime unless it has a seasoned team of technicians with a broad foundation of training and experience to maintain these highly technical systems.

Mission Critical Operations is the name of a new project being funded in round three of a U.S. Department of Labor Trade Adjustment Assistance Community College and Career Training grant.

To provide a counterpart to the systems integrator who engineers the system, an operational technology team does the day-to-day upkeep and is essentially the technology's first responders when something goes awry. If that technology happens to be a vital part of your business model, then that operational technology is mission critical, and these technicians are on the front lines. These technology personnel must be highly skilled and well trained to perform under pressure.

Mission Critical Operations (MCO) is the name of a new project being funded in round three of a U.S. Department of Labor Trade Adjustment Assistance Community College and Career Training grant. The MCO project will develop a career pathway to address demand for a mission-critical workforce able to anticipate, prevent, mitigate, and respond to mission-critical breaches. Cleveland Community College (CCC), located in Shelby, N.C., is leading an interstate consortium of schools to develop the MCO program that will produce a variety of offerings. Students ranging from those who are completely new to the field to those working in the field who need complementary or updated skills will benefit from a set of stackable and latticed articulated competencies. In addition to the academic offerings, new industry certifications will be developed. Consortium members are Nash Community College (N.C.), Wake Technical Community College (N.C.), Moultrie Technical College (Ga.), and The University of North Carolina at Charlotte. Industry partners in the project include ISA, the Automation Federation, 7x24 Exchange Carolinas, and numerous local employers.

The MCO program will combine course work from a number of traditional operational technology and information technology programs. The project will be developed and deployed over the next three years and will prepare students to handle a variety of situations in infrastructure maintenance; industrial cybersecurity; supervisory control and data acquisition systems; data analytics; automation; heating, ventilating, and air conditioning; industrial management; emergency preparedness; disaster recovery; cloud computing; telepresence; and more. Programs will be available on campus and via distance education from member institutions.

For more information, please contact the author at his email below.

ABOUT THE AUTHOR

Mitchell Sepaugh (sepaugh@clevelandcc.edu) was an OEM control systems integrator before joining CCC, where he now serves as the project manager for mission-critical operations – industrial systems and as department chair for industrial services.

New ISA84 technical report on security countermeasures related to SIS

The ISA84 standards committee on functional safety has completed and approved ISA-TR84.00.09-2013, *Security Countermeasures Related to Safety Instrumented Systems (SIS).* The new technical report provides guidance on countermeasures for reducing the likelihood of a security breach that would degrade the ability of the SIS to perform its functions.

The technical report describes performance criteria to guard against internal and external security threats to the SIS and gives guidance on how to comply with IEC 61511 and ANSI/ISA-84.00.01-2004, *Functional Safety: Safety Instrumented Systems for the Process Industry Sector,* with respect to cybersecurity.

The underlying premise of the technical report is that the means to implement, operate, and maintain system security should not compromise the performance of the SIS. To this end, SIS installations should be designed and maintained using the foundational requirements found in IEC 61511 and ANSI/ISA-84.00.01-2004, and in the ISA/IEC 62443 series on cybersecurity developed by ISA99.

For ease of use, the technical report is organized into sections that correspond to the clause numbers of ANSI/ISA-84.00.01-2004/IEC 61511-1 as follows:

- Management of functional safety
- Hazard and risk analysis
- Allocation of safety functions to protection layers
- Safety requirements specification for the safety instrumented system
- Design and engineering of the safety instrumented system
- Design and development of other means of risk reduction
- Installation, commissioning, and validation
- Operation and maintenance
- Modification
- Decommissioning

For information about viewing or obtaining this technical report or other ISA standards and technical reports, visit www.isa.org/findstandards.

IEC TC65 plenary set for Hamburg, Germany

The International Electrotechnical Commission (IEC) will hold the plenary meeting of IEC TC65, Industrial Process Measurement, Control and Automation in Hamburg, Germany, from 24 March – 4 April 2014. The first week will be devoted to working group meetings, followed by a second week of meetings of TC65 and its main subcommittees, which are:

- SC 65A, System Aspects
- SC 65B, Measurement and Control Devices
- SC 65C, Industrial Networks
- SC 65E, Devices and Integration in Enterprise Systems

ISA has had a long and productive collaborative relationship with IEC TC65, whereby a number of ISA original standards have become IEC standards, ensuring worldwide recognition and usage. These include:

- IEC 61511 series: Process Safety (ISA-84)
- IEC 61512 series: Batch Control (ISA-88)
- IEC 62264 series: Enterprise Control-System Integration (ISA-95)
- IEC 62443 series: Industrial Automation and Control Systems Security (ISA-99)
- IEC 62734: Wireless Systems for Industrial Automation: Process Control and Related Applications (ISA-100)

For more information about the IEC TC65 plenary, visit www.iec.ch.

ISA instrumentation symbols software for use with Visio

ANSI/ISA-5.1-2009, *Instrumentation Symbols and Identification,* is one of ISA's most widely used standards. It establishes a uniform means of depicting and identifying instruments or devices and their inherent functions, instrumentation systems and functions, and application software functions used for measurement, monitoring, and control. The standard presents a designation system that includes identification schemes and graphic symbols.

A new ISA software product is available for using the symbols within the Microsoft Visio software platform. It has graphic symbols for engineering design and drawing layout for a variety of instrumentation-related activities, including:

- Piping and instrument diagrams
- Process flow diagrams
- Logic diagram development
- Electrical schematic layouts
- Functional block diagrams
- Instrument loop diagrams

The software features a shape and stencil configuration for ease of symbol selection by symbol group.

The software features a shape and stencil configuration for ease of symbol selection by symbol group. It also has preassigned shape data prompts for assigning ISA-style tag numbers (prefix, functional ID, loop number, suffix) to instrument bubbles, and it has instrument bubbles formatted to accommodate long tag numbers and legend tables formatted for engineering legend sheet layout.

The new software complements a previous version intended for use within the Autodesk AutoCAD software platform.

For more information about ISA software, visit www.isa.org/software. For information about viewing or obtaining ANSI/ISA-5.1-2009, visit www.isa.org/findstandards. Select "5" from the first drop-down list and scroll down.

Focus on temperature

Temperature switch and transmitter



The PTS02 is an all-in-one temperature transmitter available with an analog 4–20 mA output, transistor, or relay switch output, and with an optional resistance temperature detector output. It uses the company's MIST technology. Temperature range, current output, switch set point, switching hysteresis, switch logic, damping, and other features are all field programmable via an optional PC interface module and software. Various sizes, configurations, and options cover a range of requirements.

Intempco, www.intempco.com

SI23 pyrometer



With laser sighting, the Sirius SI23 pyrometer has high-resolution fixed-focus optics with small spot size, digital RS232 or RS485 communication interfaces, analog 4–20 mA output, and 5 ms response speed. Optical and electronic parts are enclosed in a stainless-steel housing. Using a short wavelength spectral response of 2 to 2.6 u, the SI23 is suitable for induction heating, steel and metal heat treat processes, kilns, vacuum furnaces, research and development applications, and welding and composite temperatures above 50°C. As standard, the SI23 is offered with PSCWin software for recording, analysis, and adjustment of all pyrometer parameters. Accessories include a water-cooling jacket, air purge, scanning mirror, adjustable mounting bracket, and swivel-base mount.

Process Sensors, www.processsensorsir.com

Infrared detector



The VR40 infrared (IR) detector is a pocket-sized 4:1 IR thermometer with NCV detector and flashlight for technicians, contractors, and electricians. Its integral infrared thermometer is a safe and reliable way to measure surface temperatures, so it is suitable for heating, ventilation, air conditioning, and refrigeration professionals who need to determine the temperature of heating or cooling coils, or to detect overload currents in motors, electrical conduits, and junction boxes. Temperature readings are displayed in Fahrenheit or Celsius (–4° to 626°F or –20° to 330°C) on a four-digit LCD and are automatically held for 15 seconds. There are four sensitivity levels to optimize voltage detection over four practical ranges (12 to 25 VAC, 70 to 125 VAC, 150 to 240 VAC, 250 to 600 VAC). Detecting the presence of 12 VAC without direct contact comes in handy when troubleshooting branch circuits, as well as when working with process plant or industrial automation systems and equipment and hardwired thermostats.

General Tools, www.generaltools.com

Temperature sensor



The TS530 temperature sensor has an integrated RTD, and it combines the display, process connection, and RTD in a single part for fast and reliable performance. Push-button programming and large LED displays ensure easy operation, while a display that can turn up to 340° allows for flexible viewing in the field. The sensor also sends feedback to a programmable logic controller, allowing operators to easily monitor measurement performance from most locations. Users can mount the sensors directly to a tank or pipe without a mounting bracket using a ½ NPT process connection. For operating in harsh manufacturing environments, the sensor meets IP69K protection ratings and operates in temperatures ranging from –50° to 150°C.

TURCK, www.turck.us

Diagnostic process calibrator



Quickly find hidden loop problems with a multifunction calibrator with built-in troubleshooting tools. The PIE Model 830 uses the internal power supply to drive the loop while finding ground faults and leakage current due to corrosion or condensation in your conduits and junction boxes. It finds sensor problems by detecting two-, three-, or four-wire RTDs. The continuity beeper locates problems in loop wiring without a multimeter. The compact 830 includes a backlit display, protective rubber boot with built-in stand, hands-free carrying case, and test leads. The double-click menu system allows for quick setup, fast switching between functions, and preprogramming of set points and step/ramp times.

Practical Instrument Electronics, www.piecal.com

Precision temperature scanner



The 1586A Super-DAQ precision temperature scanner has up to 40 analog input channels and scan rates of 10 channels per second. The scanner is suitable for thermal mapping, process sensor calibration, quality-control testing, life-cycle testing, process monitoring, and environmental testing in various industries, including pharmaceutical, biotechnology, food processing, aerospace, and automotive. With both internal and external input modules, the 1586A can be used on the factory floor, where channel count and scan speeds are important, and in the calibration laboratory, where accuracy and quick input connections are required. It can measure thermocouples, platinum resistance thermometers (PRTs), thermistors, DC current, DC voltage, and resistance.

Fluke Calibration, us.flukecal.com

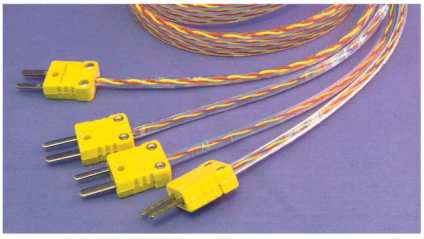
Loop-powered signal conditioner



The Jumpflex line of signal conditioners and relays now includes a loop-powered signal conditioner with conversion, isolation, and the transmission of a variety of signal types. The 857–450 Jumpflex supplies power by the output signal loop, so no separate 24 VDC power supply is required. The signal conditioner ensures safe electrical isolation of circuits at 2.5kV with less than 0.1 percent transmission error. With an extended temperature range of –25°C to 70°C, the device can be used for electrical isolation of active I/O modules from the fieldbus level. The signal conditioner offers a range of input signal types with dip-switch configuration, zero and span adjustment, and vibration-proof connections.

WAGO, www.wago.us.

Thermocouple cables



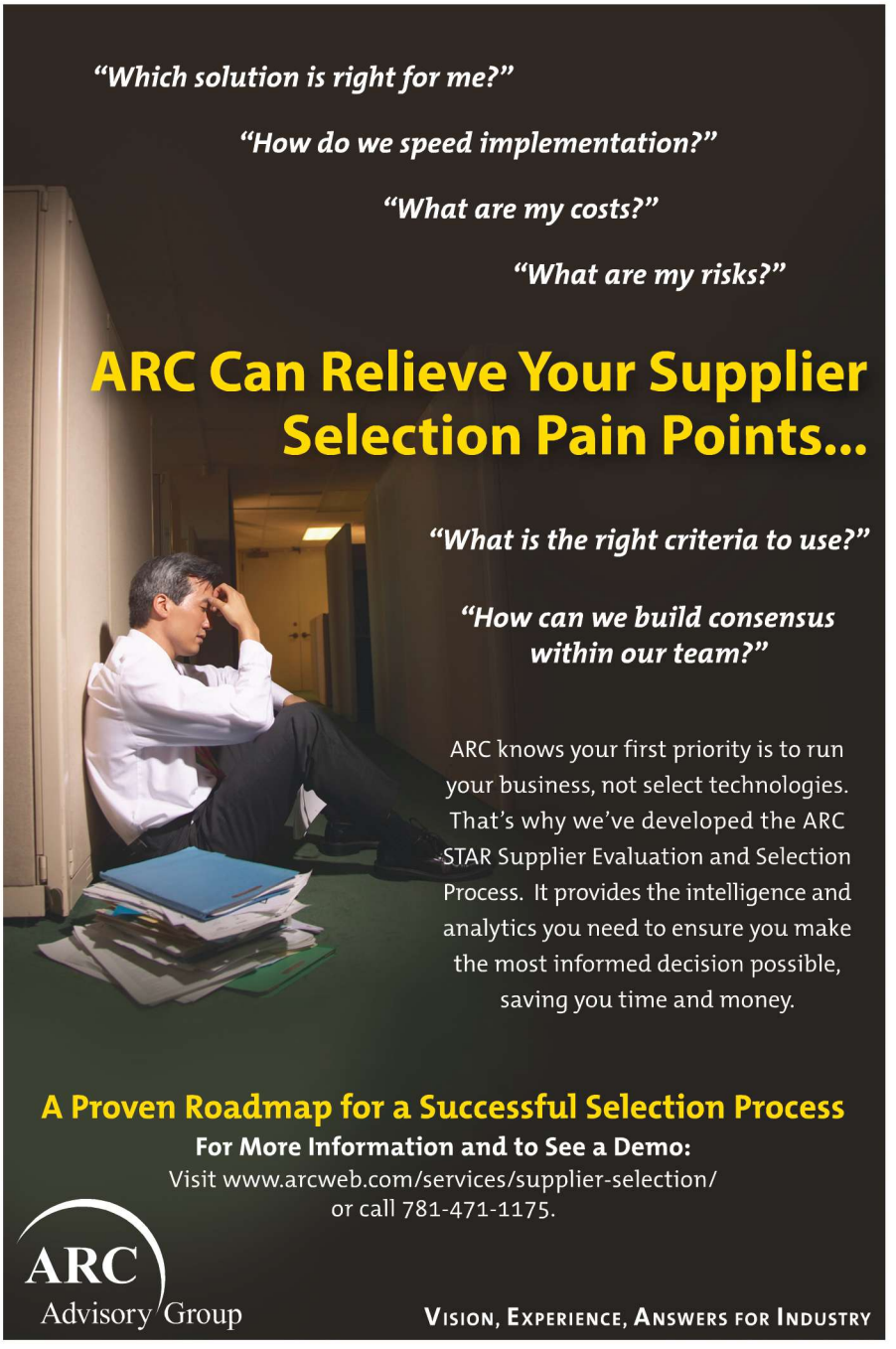
The company's flexible thermocouple cables are designed for continuous motion, tight routing, excessive temperatures (–65°C to +260°C), and extremely harsh environments. The flexible 28 AWG flat thermocouple cable is available in type J and K versions with one-to-eight conductors, and is suitable for temperature sensing, control instrumentation, heating systems, and automated equipment applications. The compact, flame-retardant solution is free of halogens and contaminants. The Flexx-Sil jacket is self-healing from small punctures and will not wear, crack, or deform from long-term exposure to tight routing, vibration, heat, continuous flexing, water, ice, steam, sunlight, humidity, ozone, UV light, autoclave, and many chemicals. Cable assemblies, complete with connectors, are offered in 3-foot, 6-foot, and 12-foot lengths.

Cicoil, www.cicoil.com



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Sample of Jobs Available at ISAJobs.org

See more at ISAJobs.org, where you can search for available jobs or advertise positions available within your company. ISA Members post resumes at no charge.

Maintenance Technician

NOVA Chemicals, Inc.: The Painesville, Ohio, plant is seeking a multifunctional maintenance technician. The technician will be responsible for performing routine service, repairs, calibration, component fabrication, and replacement and installation work on instrumentation-related components and systems in process measurement and control services. The technician will also perform and document instrumentation component and system inspections, tests, and calibrations by various means and will utilize inspection findings to determine repair requirements. Qualified applicants will have a two-year technical school diploma with three-to-five years related experience or five years combined post-secondary technical education and related experience. Mechanical or electrical maintenance skills and distributed control system experience are preferred, with ABB systems a plus . . . see more at ISAJobs.org.

Industrial Controls System Specialist

Penumbri: The company has a potential opportunity for a control systems specialist to support a government customer in Northern Va. This position involves the analysis of legacy and contemporary automation and control systems, networks, and protocols and engineering and design of control systems and networks including layered, "defense-in-depth" security architectures. The specialist will perform critical analysis of information and cybersecurity threats and recommend and implement effective countermeasures. This position is available upon contract award. The successful candidate will become a member of a five-to-seven person team to perform approximately 16 assessments of military installations per year. This position requires a bachelor's degree or applicable relevant experience. The successful candidate will have five years directly related experience in the described areas. Experience in one or more of the following is preferred: site assessment and analysis; regional, dependency and or threat analysis; remediation/mitigation planning; data development. This position requires a DoD secret clearance . . . see more at ISAJobs.org.

Electrical/Automation/Control Engineer

The National Renewable Energy Laboratory (NREL): NREL is the primary laboratory for research, development, and deployment of renewable energy technologies in the U.S. This position supports research to develop, improve, integrate, scale up, model, and demonstrate biochemical biomass conversion unit operations involved in converting lignocellulosic biomass (fibrous plant matter) to liquid transportation fuels or chemical intermediates. The engineer is responsible for improving automation of current systems and for overall control system maintenance and evolution. The successful candidate can design and program control systems and troubleshooting electrical and control system problems, perform all work in accordance with stringent safety and environmental standards, and promote a safe, supportive working environment and a culture of continuous improvement. A relevant Ph.D. or equivalent relevant education and experience is required . . . see more at ISAJobs.org.

Instrumentation Tech

City of Durham Water Management: The department seeks a congenial and skilled tradesman to complete installation, maintenance, and repair of electronic and electrical equipment, including plant control systems and motor control circuitry. The technician must be able to troubleshoot, repair, configure, and program automated control systems involving VFDs, flow/pressure/temperature monitors, process analyzers, or SCADA systems. With safety as a priority, the successful candidate must have knowledge of occupational hazards and safety precautions. He or she must be available for rotating emergency call duty. The position requires a high school diploma or equivalent; vocational courses in electronics or the electrical field are preferred. A minimum of two years of experience is required . . . see more at ISAJobs.org.

Marketing/Sales Manager

Ronan Engineering: The company seeks an experienced sales manager with the right balance of technical knowledge and demonstrated sales leadership talent. The successful candidate will thrive on competition and leadership. He or she must be able to absorb a thorough, in-depth understanding of the entire Ronan Engineering product line and to demonstrate equipment to our representatives. The individual will be expected to motivate and leverage the sales teams and strengthen communication with outside entities such as alarm annunciators, sequence-of-event recorders, and density and level systems. The candidate must have experience with SCADA, PLCs, DCSs, network control systems, TCP/IP, DNP3, and Modbus protocol. In addition, the candidate must have at least three years of experience in a process control environment and a very strong sales background with software-driven products . . . see more at ISAJobs.org.

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Urgent need for automation systems cyberprotection

By Lt. Gen. (Ret.) Robert Elder



Widespread global awareness of threats to information systems (IS) has led government and business to focus significant attention and resources on IS cybersecurity. The same cannot be said regarding industrial automation systems, where there is an urgent need to focus on the cyberprotection of critical industrial control systems.

The U.S. Department of Homeland Security has been a thought leader in this area. Its cyber ecosystem concept calls for a comprehensive approach to protect critical infrastructure going beyond traditional network and information security methodologies. The ecosystem links five activities: prevention, detection, response, recovery, and information sharing. Prevention includes built-in security, risk-based data management, and the use of trusted spaces. Detection and response form a dynamic defense to monitor behaviors and respond to potential attacks with automated defensive actions. After responding to an attack, ecosystem recovery processes execute largely automated actions to restore essential capabilities. All these activities are tied together through internal and external automated information sharing.

Although the potential impact of cyberattacks, such as Stuxnet and Idaho National Lab's experimental destruction of a power generator, is known through news stories, it still has not garnered significant attention from policymakers or industry. A recent workshop held at the Cyber Innovation Center in Bossier City, La., found that professionals find it difficult to envision the implications of an automated system protection failure. Key decision makers prefer to expend limited resources on attack prevention. Most believe that money spent in other areas detracts from this priority, and it is not necessary if the preventive measures are successful. This perception is difficult to change. Most threats are defined in terms of their attack vectors, and security professionals are very familiar with the commercial solutions designed to defeat these attacks. This is a one-dimensional understanding of the problem. Another view is to assess the value of potential targets (in military parlance, centers of gravity) or to analyze the likely intended effects of attacks from a mission or business process perspective. The former lends itself to a variety of proactive defense approaches, while the effects view is the basis for developing resiliency processes to limit the effectiveness of attacks. Commercial products are available to support both approaches, but their capabilities are not widely known among cybersecurity professionals.

Addressing cyberprotection requires a sense of urgency among cybersecurity, industry, and government leaders. Proactive defense and resiliency solutions require extensive coordination between these groups. Systems maintenance and security professionals must develop a better understanding of the business lines they support, and business executives must better understand the challenges of operating automated systems in contested environments.

Workshop participants coalesced around several key recommendations. First, expect cyberspace to be degraded: design processes to remain effective when bandwidth is limited. Second, balance system maintenance ease with diversity and redundancy to enhance survivability and build recovery capacity. Third, implement rules to reduce network noise so detection processes can operate more effectively. Fourth, leverage inherent resiliency opportunities: integrate protective measures across the operational, logical, physical, and infrastructure networking levels. Fifth, provide a means to insert human decision making in automated response and recovery control loops. Finally, develop a risk management approach that balances resource allocations across the entire cyber ecosystem: protection, detection, response, and recovery.

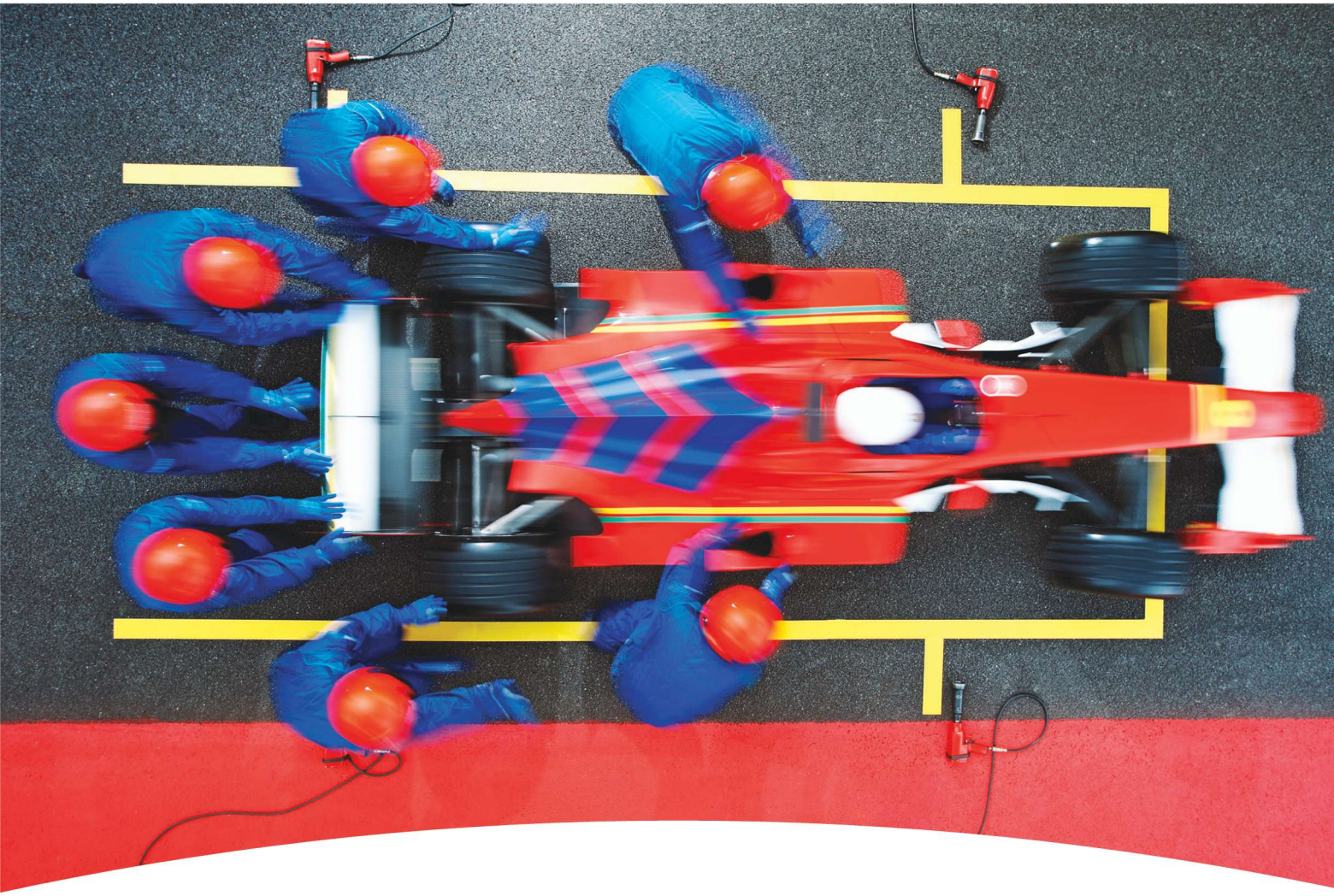
This takes teamwork! Everyone involved has a critical role in the protection of industrial automation systems. Developers must eliminate vulnerabilities with a combination of hardware controls and software assurance. Threat analysts must seek information on attack vectors and develop a situational understanding of the intentions and behaviors of potential threat actors. Network and process designers must demand resiliency and diversity among critical systems, implementing controls and audits to detect potential issues before they become crises. Finally, operators of automated systems must implement business processes that support the professionals that maintain and secure these systems. Leadership is critical to implement these cultural changes.

Action is essential. Fortunately, there are many organizations available to provide assistance, including the Cyber Technology and Information Security Laboratory at the Georgia Tech Research Institute that uses expertise in systems engineering, signals, and other technology areas to create resilient control solutions for operations in contested environments and to help industry safeguard the nation's critical infrastructure. ISA's cybersecurity standards and programs are also a valuable resource.

ABOUT THE AUTHOR

Lieutenant General Robert Elder (USAF, retired) (relder@gmu.edu) joined the George Mason University faculty as a research professor with the Volgenau School of Engineering following retirement from the Air Force. He also serves as a senior advisor to the Georgia Tech Research Institute and the Cyber Innovation Center. Elder was the first commander of Air Force Network Operations and led the development of the cyberspace mission for the Air Force. He holds a doctorate of engineering from the University of Detroit.

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